MT220 · Week 2 — Central Tendency & Dispersion

Complete study pack: notes for slides 1–49 (10 per topic) · fully worked answers. All formulas follow the lecture slides.

Part A — Measures of Central Tendency

A **measure of central tendency** is one number that represents the whole dataset — its "middle" or most typical value. Three main measures: **Mean** (average), **Median** (middle value), **Mode** (most frequent value).

1. Arithmetic Mean (Slides 6–8)

Data type	Formula	Slide example
Ungrouped (raw)	$\bar{X} = \Sigma X / n$	10, 15, 20, 25, 30 → 100/5 = **20**
Discrete frequency	$\bar{X} = \Sigma fX / \Sigma f$	X 10, 20, 30; f 2, 5, 3 → 210/10 = **21**
Grouped (classes)	$\bar{X} = \Sigma fx / N$, x = midpoint	Running example → 745/30 = **24.83**

- **Midpoint** x = (Lower limit + Upper limit) / 2. We assume all values of a class sit at its midpoint.
- Classes 1–10, 11–20 … are **inclusive** (they have a gap), so their true **class boundaries** are 0.5–10.5, 10.5–20.5 … Boundaries are used for median, mode and quartiles — **not** for the midpoint.
- Excel: `=AVERAGE(A1:A5)` ; frequency mean `=SUM(fx)/SUM(f)` .

Running example (used all lecture) — marks of 30 students:

Class	Boundary	f	x	fx	c.f.
1–10	0.5–10.5	4	5.5	22	4
11–20	10.5–20.5	6	15.5	93	10
21–30	20.5–30.5	10	25.5	255	20
31–40	30.5–40.5	8	35.5	284	28
41–50	40.5–50.5	2	45.5	91	30
Total		**30**		**745**	

Results: Mean **24.83** · Median **25.5** · Mode **27.17** · Q_1 **16.33** · Q_3 **33.63** · IQR **17.29** · QD **8.65** · σ^2 **126.22** · σ **11.24**

2. Weighted Mean (Slide 9)

$\bar{X}w = \Sigma WX / \Sigma W$ — used when values differ in importance (credit hours, quantities).
Slide: Maths 80 (4 cr), English 70 (2), Statistics 90 (3) → 730/9 = **81.1** (simple mean = 80).
If all weights are equal → weighted mean = simple mean.

3. Median (Slides 11–13)

- **Raw data:** sort ascending. Odd n → middle value; even n → average of the two middle values. (4, 7, 9, 10, 15 → **9**; 3, 4, 5, 8, 10, 12 → (5+8)/2 = **6.5**)
- **Discrete frequency:** find n/2, then the **first c.f. ≥ n/2**. (n = 12, n/2 = 6, first c.f. ≥ 6 is 10 at X = 30 → Median = **30**)
- **Grouped: Median = L + [(N/2 − cf) / f] × c**
  L = lower **boundary** of the median class · cf = c.f. **before** it · f = its frequency · c = class width.
  Running example: N/2 = 15 → class 21–30; Median = 20.5 + [(15 − 10)/10] × 10 = **25.5**

4. Mode (Slides 15–16)

- **Raw:** the most frequent value. One mode = unimodal; two+ = bimodal/multimodal; all different = no mode. Excel `=MODE.SNGL` .
- **Grouped: Mode = $L + [(f_1 - f_o) / (2f_1 - f_o - f_2)] \times c$**
  f_1 = frequency of modal class (highest f), f_o = class before, f_2 = class after (use 0 if none).

Running example: 20.5 + [(10 − 6)/(20 − 6 − 8)] × 10 = 20.5 + 6.67 = **27.17**

5. Empirical Relationship (Slide 17)

Mode = 3 × Median − 2 × Mean (moderately skewed data only; an approximation).
Slide: Mean 20, Median 18 → Mode = 54 − 40 = **14**.

6. Quartiles (Slides 19–21)

Quartiles split ordered data into 4 equal parts: $\mathbf{Q_1}$ (25% below), $\mathbf{Q_2}$ **= Median** (50%), $\mathbf{Q_3}$ (75%).

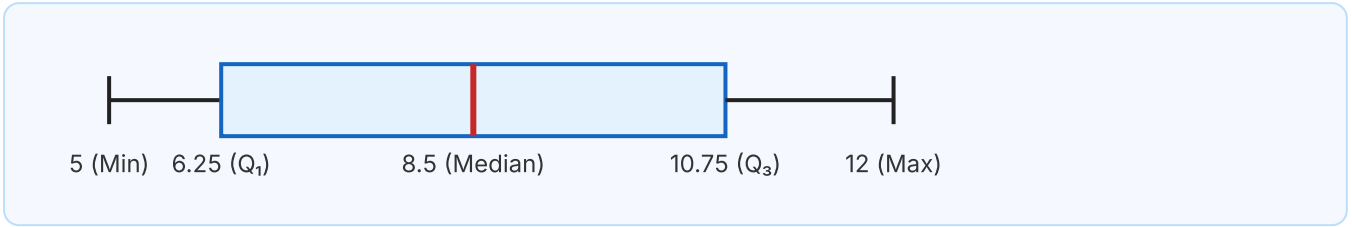
- **Raw data:** positions $Q_1 = (n+1)/4$, $Q_2 = 2(n+1)/4$, $Q_3 = 3(n+1)/4$, then interpolate.
  Slide: 5, 6, 7, 8, 9, 10, 11, 12 → $Q_1 = 6 + 0.25(1)$ = **6.25**, Q_2 = **8.5**, Q_3 = **10.75**. Excel `=QUARTILE.INC(range, k)`.
- **Grouped:** $\mathbf{Q_k = L + [(kN/4 - cf) / f] \times c}$
  Running example: Q_1: N/4 = 7.5 → 10.5 + [(7.5 − 4)/6] × 10 = **16.33**; Q_3: 3N/4 = 22.5 → 30.5 + [(22.5 − 20)/8] × 10 = **33.63**

7. IQR and Quartile Deviation (Slide 22)

IQR = Q_3 − Q_1 — spread of the middle 50%; less affected by extreme values than the range.
QD = IQR / 2 (semi-interquartile range).
Running example: IQR = 33.63 − 16.33 = **17.29** (17.30 with rounded Q's); QD = **8.65**.

8. Box Plot (Slide 23)

Five-number summary: **Min, Q_1, Median, Q_3, Max**. The box = middle 50% (IQR), line inside = median, whiskers to min and max; points beyond whiskers = possible outliers.
Slide: 5, 6, 7, 8, 9, 10, 11, 12 → Min 5, Q_1 6.25, Median 8.5, Q_3 10.75, Max 12.



9. Applications (Slide 25)

- AI: mean accuracy across cross-validation folds; **median** inference time when a few requests spike (outliers).
- Cyber: mean/median response time = performance baseline; **mode** = most common alert type in logs.

Part B — Measures of Dispersion

Dispersion (variation / spread) = how far values are scattered around the mean. Same mean can hide very different spreads: Class A 45–49 (mean 47, consistent) vs Class B 20–80 (mean 50, inconsistent). Measures: Range, IQR, Variance, S.D., C.V.

10. Range and Coefficient of Range (Slide 28)

Range = L − S (largest − smallest). Quick, but uses only the two extremes.
Coeff. of Range = (L − S) / (L + S) — unit-free, for comparing datasets.
Slide: 150 ... 170 cm → Range = **20 cm**; Coeff. = 20/320 = **0.0625 = 6.25%**.

11. Variance and Standard Deviation (Slides 29–31)

Data	Variance σ^2	S.D. σ
Ungrouped	$\Sigma(X - \bar{X})^2 / n$	$\sqrt{[\Sigma(X - \bar{X})^2 / n]}$

Frequency (shortcut)	$\Sigma fx^2/N - (\Sigma fx/N)^2$	$\sqrt{[\Sigma fx^2/N - (\Sigma fx/N)^2]}$

- Variance is in **squared units**; S.D. is back in the original units. **Smaller S.D. → more consistent.**
- Slide (ungrouped): 2, 4, 6, 8, 10 → $\Sigma(X - \bar{X})^2 = 40 \rightarrow \sigma^2 = \mathbf{8}$, $\sigma = \mathbf{2.83}$.
- Slide (grouped): $22287.5/30 - (745/30)^2 = 742.92 - 616.69 = \mathbf{126.22}$, $\sigma = \mathbf{11.24}$.
- ⚠ $fx^2 = f \times x^2$ (e.g. $4 \times 5.5^2 = 121$) — **not** $(fx)^2$. ⚠ Check: $\Sigma(X - \bar{X})$ must equal 0.

12. Coefficient of Variation (Slides 32–33)

C.V. = $(\sigma / \bar{X}) \times 100\%$ — relative spread; compares datasets with different units or means. **Lower C.V. = more consistent.** Slide Example 4.9: Price C.V. = **41.62%**, Life C.V. = **37.15%** → Price varies more, even though Life's S.D. (78.38) is much larger than Price's (7.66).

13. Applications of Dispersion (Slide 35)

- AI: low variance across folds = stable model; high variance = overfitting (bias–variance trade-off).
- Cyber: **3-sigma rule** — flag values beyond mean ± 3σ; a DDoS attack sharply increases variance of request rates.

Part C — Skewness and Kurtosis

14. Skewness (Slides 37–38)

Skewness = **lack of symmetry**.

Type	Tail	Order	Sk
Symmetric	—	Mean = Median = Mode	0
Positive (right)	long right tail	Mean > Median > Mode	> 0
Negative (left)	long left tail	Mean < Median < Mode	< 0

Karl Pearson: **Sk = (Mean − Mode) / σ**, or **Sk = 3(Mean − Median) / σ** when the mode is unstable.
Running example: (24.83 − 27.17)/11.24 = **−0.21**; 3(24.83 − 25.5)/11.24 = **−0.18** → slightly negatively skewed.

15. Kurtosis (Slides 39–40)

Kurtosis = peakedness / tail-heaviness compared with Normal. **$\beta_2 = [\Sigma f(X - \bar{X})^4/N] \div [\Sigma f(X - \bar{X})^2/N]^2$** = m_4 / m_2^2.

Type	β_2	Excess ($\beta_2 - 3$)	Shape
Mesokurtic	= 3	= 0	Normal
Leptokurtic	> 3	> 0	Peaked, fat tails, more outliers
Platykurtic	< 3	< 0	Flat, fewer/milder outliers

⚠ Excel `=KURT` gives **excess** kurtosis — compare with 0, not 3.

16. Applications (Slide 41)

- Right-skewed features (income, file size) → log-transform before training.
- High kurtosis → outliers more frequent than Normal predicts (fraud detection uses it).
- Response times are right-skewed — assuming Normal can create false anomalies.

Part D — Excel (Slides 43–45)

Measure	Function	Slide result (12 marks)
Mean / Median / Mode	=AVERAGE · =MEDIAN · =MODE.SNGL	61.08 · 60 · 60
Range	=MAX − MIN	82 − 45 = 37

Q_1 / Q_3	=QUARTILE.INC(range,1 / 3)	52.75 / 68.75
Sample variance / S.D.	=VAR.S · =STDEV.S (÷ n − 1)	117.36 · 10.83
Population variance / S.D.	=VAR.P · =STDEV.P (÷ n)	107.58 · 10.37
C.V.	=STDEV.S(...)/AVERAGE(...)*100	17.73%
Skewness / Excess kurtosis	=SKEW · =KURT	0.45 · −0.14

Sample (part of the data) → divide by **n − 1**. Population (all of the data) → divide by **n**. Hand formulas in this lecture use **n**.

Formula Sheet

Mean	$\bar{X} = \Sigma X/n \cdot \Sigma fX/\Sigma f \cdot \Sigma fx/N$
Weighted mean	$\bar{X}w = \Sigma WX / \Sigma W$
Median (grouped)	$L + [(N/2 - cf)/f] \times c$
Mode (grouped)	$L + [(f_1 - f_o)/(2f_1 - f_o - f_2)] \times c$
Empirical	Mode = 3 Median − 2 Mean
Quartiles	raw: position $k(n+1)/4$ · grouped: $L + [(kN/4 - cf)/f] \times c$
IQR / QD	$Q_3 - Q_1 \cdot (Q_3 - Q_1)/2$
Range / Coeff.	$L - S \cdot (L - S)/(L + S)$
Variance	$\Sigma(X - \bar{X})^2/n \cdot \Sigma fx^2/N - (\Sigma fx/N)^2$
C.V.	$(\sigma / \bar{X}) \times 100\%$
Skewness	(Mean − Mode)/σ · 3(Mean − Median)/σ
Kurtosis	$\beta_2 = m_4/m_2^2$ · excess = $\beta_2 - 3$

Top Mistakes to Avoid

1. Using class **limits** instead of **boundaries** for L in median/mode/quartiles.
2. Using the c.f. **of** the median class instead of the c.f. **before** it.
3. Forgetting to **sort** raw data before finding the median or quartiles.
4. Writing $(fx)^2$ instead of $f \times x^2$.
5. Forgetting the square root (giving the variance as the S.D.).
6. Comparing S.D.s of datasets with different means/units — use **C.V.**
7. Comparing Excel KURT with 3 — it is already excess kurtosis (compare with 0).
8. Using f_o or f_2 from the wrong classes in the mode formula (0 if no class exists).

Part E — Exam-Style Question: Grouped Frequency Distribution & Box Plot

Your teacher's paper question gives **raw data** and asks you to (a) build the table, (b) find mean/median/mode, (c) Q_1 and Q_3, (d) draw a box plot, (e) comment on the shape. Total 25 marks. Learn this **exact routine**.

Step-by-step routine (memorise)

1. **Sort / scan the data.** Note the **minimum** and **maximum** (you need them for the box plot).
2. **Write the classes.** Start where told (e.g. 31–50). Upper limit = lower + width − 1. Keep adding the width until the last class **contains the maximum**.
3. **Class boundaries** = lower limit − 0.5 and upper limit + 0.5 (because there is a gap of 1 between classes).
4. **Tally:** go through the data once, putting a stroke in the right class; every 5th stroke crosses the previous four (𝍸). Then count → **f**.
5. **Check Σf = n** (number of observations). If not, you missed or double-counted a value.
6. **Midpoint** x = (lower limit + upper limit) / 2.
7. **Cumulative frequency** c.f. = running total of f; the last c.f. must equal N.
8. **(b) Mean** = Σfx/N. **Median**: N/2 → median class → $L + [(N/2 - cf)/f] \times c$. **Mode**: highest f → $L + [(f_1 - f_o)/(2f_1 - f_o - f_2)] \times c$. Always **name the median class and modal class** — they carry marks.
9. **(c)** Q_1 at N/4, Q_3 at 3N/4 with $L + [(kN/4 - cf)/f] \times c$.
10. **(d) Box plot:** draw a **scale** first, then the box from Q_1 to Q_3, a line at the median, whiskers to Min and Max. **Label all five values.**
11. **(e) Shape:** compare Mean, Median, Mode **and** use the box plot (longer whisker side, median position inside the box). Write 2–3 sentences.

Shape rules for (e)

Shape	Order	Box plot clue
Positively (right) skewed	Mean > Median > Mode	Right whisker longer; median nearer Q_1
Negatively (left) skewed	Mean < Median < Mode	Left whisker longer; median nearer Q_3
Symmetric	Mean ≈ Median ≈ Mode	Whiskers similar; median in middle of box

Marks-losing mistakes

- Last class not reaching the maximum (167 needs class 151–170).
- Using class limits instead of boundaries for L.
- Forgetting $f_o = 0$ when the modal class is the first class.
- Using the grouped Q_1/Q_3 but forgetting the box plot needs the **actual** Min and Max from the raw data.
- Box plot without a scale or without labels.
- Saying "skewed" without saying **which direction** and **why**.

Teacher's Question 1 — Fully Solved

Question 1 — Grouped Frequency Distribution & Box Plot. The response times (ms) for 50 requests to a university web server:

```
40   41   41   41   42   43   43   43   44   44   46   48   49   49   49   49   49   51
51   52   54   55   56   57   59   65   67   68   68   70   71   72   74   75   76   77
80   82   83   86   87   97   98   98   102   106   109   118   150   167
```

(a) Construct a grouped frequency distribution table using a class width of 20 starting from the class 31–50. Show class limits, class boundaries, tally/frequency, midpoint and cumulative frequency. *[7]*
(b) Calculate the Mean, Median and Mode (show the median/modal class and all working). *[6]*
(c) Calculate Q_1 and Q_3 from your table. *[4]*
(d) Construct a box plot using the five-number summary (use the actual minimum and maximum). *[4]*
(e) Comment on the shape of the distribution (symmetric, positively or negatively skewed). Justify in 2–3 sentences. *[4]*

(a) Range = 167 − 40 = 127. Classes of width 20 starting 31–50; last class 151–170 contains the maximum 167. Boundaries = lower − 0.5 and upper + 0.5; midpoint = (lower + upper)/2.

Class limits	Class boundaries	Tally	f	Midpoint x	c.f.
31 – 50	30.5 – 50.5	𝍸 𝍸 𝍸 ||	17	40.5	17
51 – 70	50.5 – 70.5	𝍸 𝍸 |||	13	60.5	30
71 – 90	70.5 – 90.5	𝍸 𝍸 |	11	80.5	41
91 – 110	90.5 – 110.5	𝍸 |	6	100.5	47
111 – 130	110.5 – 130.5	|	1	120.5	48
131 – 150	130.5 – 150.5	|	1	140.5	49
151 – 170	150.5 – 170.5	|	1	160.5	50
Total			**N = 50**		

Check: Σf = 50 = number of observations ✓

(b)

Class	f	x	fx
31 – 50	17	40.5	688.5
51 – 70	13	60.5	786.5
71 – 90	11	80.5	885.5
91 – 110	6	100.5	603
111 – 130	1	120.5	120.5
131 – 150	1	140.5	140.5
151 – 170	1	160.5	160.5
Total	**50**		**3385**

Mean = Σfx / N = 3385 / 50 = **67.7 ms**

Median: N/2 = 25 → first c.f. ≥ 25 is 30 → **median class 51 – 70** (boundary 50.5 – 70.5).
L = 50.5, cf = 17, f = 13, c = 20
Median = 50.5 + [(25 − 17) / 13] × 20 = 50.5 + 12.31 = **62.81 ms**

Mode: highest f = 17 → **modal class 31 – 50** (boundary 30.5 – 50.5).
L = 30.5, f_1 = 17, f_o = 0 (no class before), f_2 = 13, c = 20

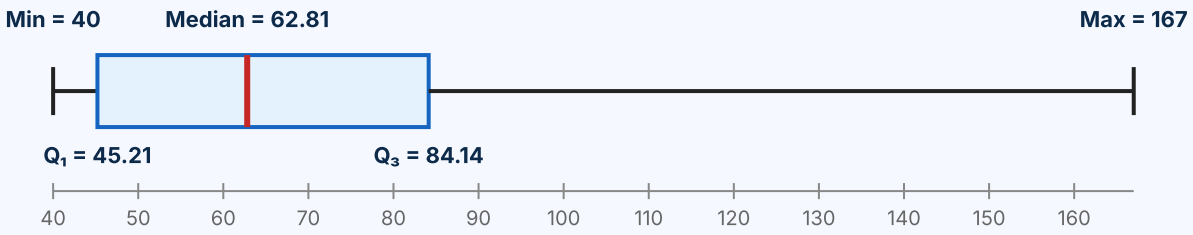
Mode = 30.5 + [(17 − 0) / (2×17 − 0 − 13)] × 20 = 30.5 + 16.19 = **46.69 ms**

(c)

Q_1: N/4 = 12.5 → class 31 – 50 (L = 30.5, cf = 0, f = 17, c = 20)
Q_1 = 30.5 + [(12.5 − 0) / 17] × 20 = 30.5 + 14.71 = **45.21 ms**

Q_3: 3N/4 = 37.5 → class 71 – 90 (L = 70.5, cf = 30, f = 11, c = 20)
Q_3 = 70.5 + [(37.5 − 30) / 11] × 20 = 70.5 + 13.64 = **84.14 ms**

(d) Five-number summary: Min = **40**, Q_1 = **45.21**, Median = **62.81**, Q_3 = **84.14**, Max = **167**. IQR = 38.93.



(e) Mean = 67.7, Median = 62.81, Mode = 46.69 → **Mean > Median > Mode**. In the box plot the right whisker is much longer than the left (left 5.21, right 82.86) and the median line lies closer to Q_1 (right half of the box is wider). Therefore the distribution is **positively (right) skewed**.

Practice Questions (100)

Try every question on paper first, then check the worked answers in the next section. Questions marked **(Slide ...)** are taken directly from the lecture.

Set 1: Arithmetic Mean

1. Find the mean of: 12, 15, 18, 20, 25.
2. Find the mean of: 45, 52, 38, 61, 49, 55.
3. Daily temperatures (°C) for a week are: 28, 31, 30, 29, 32, 33, 27.
4. Find the mean: X = 5, 10, 15, 20 with f = 3, 7, 6, 4.
5. Number of children per family: X = 0, 1, 2, 3, 4 with f = 4, 8, 10, 5, 3. Find the mean.
6. **(Slide Q2)** Find the mean: Class 1–10, 11–20, 21–30, 31–40; f = 3, 5, 7, 3.
7. Find the mean: Class 0–10, 10–20, 20–30, 30–40, 40–50; f = 5, 8, 15, 10, 2.
8. Find the mean: Class 21–30, 31–40, 41–50, 51–60, 61–70; f = 6, 10, 14, 7, 3.
9. The mean of 6 numbers is 15. Five of them are 10, 12, 14, 18, 20. Find the sixth number.
10. Section A (30 students) has mean 60; Section B (20 students) has mean 70. Find the combined mean.

Set 2: Weighted Mean

1. **(Slide Q4)** Scores 85, 90, 78 in tests worth 2, 3, 1 credit hours. Find the weighted mean.
2. Marks 75, 82, 68, 90 in subjects of 3, 4, 2, 3 credit hours. Find the weighted mean.
3. A shop sells rice at Rs 50/kg (10 kg), Rs 80/kg (5 kg), Rs 120/kg (2 kg). Find the average price per kg.
4. Grade points 4.0, 3.3, 2.7, 3.7 for courses of 3, 3, 2, 4 credit hours. Find the GPA.
5. Quiz = 20%, Mid = 30%, Final = 50%. Scores: 80, 70, 85. Find the overall score.
6. Values 60, 70, 80 all have weight 2. Find the weighted mean and compare with the simple mean.
7. Marks 80 (weight 2) and 60 (weight w) give a weighted mean of 70. Find w.
8. Values 90 (weight 1) and 60 (weight 4). Find the simple mean and the weighted mean.
9. **(AI)** A model scores 92% accuracy on 1000 test samples and 85% on 4000 samples. Find the overall accuracy.
10. **(Cyber)** Server A: average response 100 ms over 200 requests; Server B: 150 ms over 300 requests. Find the overall average response time.

Set 3: Median

1. Find the median of: 7, 3, 9, 1, 5.
2. Find the median of: 12, 5, 8, 20, 15, 10.
3. Find the median of: 23, 19, 31, 27, 19, 25, 30.
4. Response times (ms): 100, 105, 98, 250, 102, 101. Find the median. Why is it better than the mean here?
5. Find the median: X = 1, 2, 3, 4, 5; f = 3, 5, 8, 4, 2.
6. Find the median: X = 10, 20, 30, 40, 50; f = 4, 6, 5, 3, 2.
7. **(Slide Q2)** Find the median: Class 1–10, 11–20, 21–30, 31–40; f = 3, 5, 7, 3.
8. Find the median: Class 0–10, 10–20, 20–30, 30–40, 40–50; f = 5, 8, 15, 10, 2. (No gaps, so boundaries = limits.)
9. Find the median: Class 21–30, 31–40, 41–50, 51–60, 61–70; f = 6, 10, 14, 7, 3.
10. Find the median: Class 100–119, 120–139, 140–159, 160–179, 180–199; f = 8, 14, 20, 12, 6.

Set 4: Mode and the Empirical Relationship

1. Find the mode: 2, 3, 3, 5, 7, 3, 8.
2. Find the mode: 4, 6, 6, 8, 9, 9, 10.
3. Find the mode: 1, 2, 3, 4, 5.
4. Shoe sizes sold: size 6, 7, 8, 9, 10 with f = 5, 12, 20, 15, 8. Find the mode.
5. **(Slide Q2)** Find the mode: Class 1–10, 11–20, 21–30, 31–40; f = 3, 5, 7, 3.

6. Find the mode: Class 0–10, 10–20, 20–30, 30–40, 40–50; f = 5, 8, 15, 10, 2.
7. Find the mode: Class 21–30, 31–40, 41–50, 51–60, 61–70; f = 6, 10, 14, 7, 3.
8. Find the mode: Class 0–10, 10–20, 20–30, 30–40; f = 12, 8, 5, 3.
9. Mean = 50, Median = 48. Estimate the mode.
10. Mode = 30, Mean = 36. Estimate the median.

Set 5: Quartiles, IQR, QD and Box Plot

1. Find Q_1, Q_2, Q_3, IQR and QD: 3, 7, 8, 12, 13, 14, 18, 21.
2. Find Q_1, Q_2, Q_3: 2, 4, 6, 8, 10, 12, 14.
3. Find Q_1, Q_2, Q_3 and IQR: 15, 9, 22, 11, 18, 25, 13, 20, 17, 10, 14.
4. **(Slide Q3)** Find Q_1, Q_3, IQR and QD: Class 1–10, 11–20, 21–30, 31–40; f = 3, 5, 7, 3.
5. Find Q_1, Q_3 and IQR: Class 0–10, 10–20, 20–30, 30–40, 40–50; f = 5, 8, 15, 10, 2.
6. Find Q_1, Q_3, IQR and QD: Class 21–30, 31–40, 41–50, 51–60, 61–70; f = 6, 10, 14, 7, 3.
7. $Q_1 = 40$ and $Q_3 = 64$. Find the IQR and the Quartile Deviation.
8. Give the five-number summary for 3, 7, 8, 12, 13, 14, 18, 21 and describe its box plot.
9. Section A: $Q_1 = 50$, $Q_3 = 70$. Section B: $Q_1 = 45$, $Q_3 = 80$. Which section's middle 50% is more spread out?
10. For 2, 4, 6, 8, 10, 12, 14 show that Q_2 equals the median.

Set 6: Range and Coefficient of Range

1. **(Slide Q5)** Find the range and coefficient of range: 12, 18, 9, 25, 30.
2. Find the range and coefficient of range: 40, 55, 32, 70, 48.
3. Temperatures (°C): 18, 25, 30, 22, 35. Find the range and coefficient of range.
4. Prices range from Rs 250 to Rs 400. Find the coefficient of range.
5. Exam marks: lowest 33, highest 97. Find the range and coefficient of range.
6. The largest value is 95 and the coefficient of range is 0.25. Find the smallest value.
7. Response times (ms): 120, 135, 128, 410, 125. Find the range. What limitation does this show?
8. A: 10, 10, 10, 10, 50 and B: 10, 20, 30, 40, 50. Find both ranges. What does this show?
9. **(Slide 27 data)** Class A: 45, 46, 47, 48, 49; Class B: 20, 35, 50, 65, 80. Compare using coefficient of range.
10. Heights: 150–190 cm. Weights: 45–90 kg. Which varies more? (Use coefficient of range.)

Set 7: Variance and Standard Deviation

1. **(Slide Q6)** Find the variance and S.D.: 3, 6, 9, 12, 15.
2. Find the variance and S.D.: 10, 20, 30, 40, 50.
3. Find the variance and S.D.: 5, 7, 9, 11, 13, 15.
4. **(Slide 27, Class A)** Find the variance and S.D.: 45, 46, 47, 48, 49.
5. **(Slide 27, Class B)** Find the variance and S.D.: 20, 35, 50, 65, 80.
6. Find the variance and S.D.: X = 1, 2, 3, 4, 5; f = 4, 8, 10, 5, 3.
7. Find the variance and S.D.: Class 1–10, 11–20, 21–30, 31–40; f = 3, 5, 7, 3.
8. Find the variance and S.D.: Class 0–10, 10–20, 20–30, 30–40, 40–50; f = 5, 8, 15, 10, 2.
9. Find the variance and S.D.: Class 21–30, 31–40, 41–50, 51–60, 61–70; f = 6, 10, 14, 7, 3.
10. For 4, 8, 6, 2, 10 find the population variance (÷n) and the sample variance (÷(n − 1)).

Set 8: Coefficient of Variation

1. **(Slide Q7)** Machines A and B: mean 10 mm; S.D. A = 0.5 mm, B = 0.8 mm. Which is more consistent?
2. X: mean 40, S.D. 8. Y: mean 120, S.D. 18. Which is more consistent?
3. Heights: mean 160 cm, S.D. 8. Weights: mean 60 kg, S.D. 6. Which varies more?
4. Batsman A scores 45, 50, 55, 60, 40; Batsman B scores 20, 90, 35, 80, 25. Who is more consistent?
5. **(Slide Example 4.9)** Prices: 8, 13, 18, 23, 30. Life (hrs): 130, 150, 180, 250, 345. Which has greater relative variation?
6. Using your answers for Class 1–10 ... 31–40 (f = 3, 5, 7, 3), find the C.V.

7. C.V. = 25% and mean = 40. Find the S.D.
8. C.V. = 12% and S.D. = 3. Find the mean.
9. **(AI)** Model A: mean accuracy 90%, S.D. 2. Model B: mean 85%, S.D. 1.5 (across folds). Which is more stable?
10. **(Slide 27 data)** Class A: mean 47, $\sigma = 1.41$; Class B: mean 50, $\sigma = 21.21$. Compare consistency.

Set 9: Skewness

1. **(Slide Q9)** Mean = 32, Median = 30, Mode = 26, S.D. = 8. Find Sk (using mode). Type?
2. Mean = 45, Median = 48, Mode = 50, S.D. = 10. Find Sk by both formulas.
3. Mean = Median = Mode = 40, S.D. = 5. Find Sk.
4. Mean = 60, Median = 55, S.D. = 15 (mode not given). Find Sk.
5. **(Slide example)** Mean = 24.83, Median = 25.5, Mode = 27.17, S.D. = 11.24. Find Sk by both formulas.
6. Mean = 70, Median = 66, S.D. = 12. Estimate the mode, then find Sk.
7. Sk = 0.5, Mean = 40, S.D. = 8. Find the mode.
8. Mean = 50, Median = 52, Mode = 55. Without calculating, state the skewness.
9. Data: 2, 3, 3, 3, 4, 5, 9. Find the mean, median, mode, S.D. and Sk (using mode).
10. Income data usually has a long right tail. Which is larger, the mean or the median? Why?

Set 10: Kurtosis

1. **(Slide Q10)** Excel reports excess kurtosis = +2.5. Type? Outliers?
2. Find the kurtosis of 2, 4, 6, 8, 10 using $\beta_2 = m_4/m_2^2$.
3. Find the kurtosis of 1, 5, 5, 5, 5, 5, 9.
4. Excel KURT = −1.2. Type?
5. $\beta_2 = 3$. Type? What is the excess kurtosis?
6. $m_2 = 4$ and $m_4 = 40$. Find β_2 and the type.
7. $m_2 = 5$ and $m_4 = 100$. Find β_2 and the type.
8. Kurtosis $\beta_2 = 4.2$. Find the excess kurtosis and the type.
9. Dataset A: excess kurtosis 1.5; Dataset B: −0.8. Which has fatter tails?
10. **(Cyber)** Login-attempt data has kurtosis $\beta_2 = 7$. What does this suggest?

Set 11: Table Construction & Box Plot Basics (concept drill)

1. Class limits 31–50. Find the class boundaries, class width and midpoint.
2. Classes 10–19, 20–29, 30–39. Write the class boundaries.
3. Class boundaries 50.5 – 70.5. Find the class limits (integer data) and midpoint.
4. Data range 40 to 167, width 20, starting at 31–50. List all classes.
5. Make the tally and frequency for the class 41–50 from: 38, 41, 45, 50, 51, 47, 49, 44, 52, 50.
6. Frequencies 6, 11, 15, 9, 4. Write the cumulative frequencies.
7. N = 50 and c.f. are 17, 30, 41, 47, 48, 49, 50. Which class contains the median? Q_1? Q_3?
8. The modal class is the first class with f = 17; the next class has f = 13. What are f_o and f_2?
9. Five-number summary: 40, 45.2, 62.8, 84.1, 167. Which whisker is longer? What does it suggest?
10. Mean = 70.4, Median = 62.8, Mode = 46.7. State the shape and justify.

Exam-Style Practice (10 full questions, same format as the paper)

Each question has 5 parts exactly like the teacher's paper (25 marks). Solutions are in the Worked Answers section.

Exam Practice 1. Login times (seconds) for 50 users of a university portal (50 values):

1 2 3 3 4 5 5 5 6 6 6 7 7 8 9 9 9 10 10 11 11
12 12 12 12 12 13 14 15 15 16 17 17 19 19 20 23 23 23
23 24 25 26 27 29 30 30 31 32 33

(a) Construct a grouped frequency distribution table using a class width of 10 starting from the class 1–10. Show class limits, class boundaries, tally/frequency, midpoint and cumulative frequency. *[7]*
(b) Calculate the Mean, Median and Mode (show the median/modal class and all working). *[6]*
(c) Calculate Q_1 and Q_3 from your table. *[4]*
(d) Construct a box plot using the five-number summary (use the actual minimum and maximum). *[4]*
(e) Comment on the shape of the distribution (symmetric, positively or negatively skewed). Justify in 2–3 sentences. *[4]*

Exam Practice 2. Marks of 45 students in a Statistics quiz (out of 100) (45 values):

38 45 49 56 65 65 68 68 68 70 72 72 73 73 74 74 74 75
76 76 76 78 79 79 80 80 81 81 81 81 82 82 83 83 84 84
85 85 85 86 86 87 88 88 89

(a) Construct a grouped frequency distribution table using a class width of 10 starting from the class 31–40. Show class limits, class boundaries, tally/frequency, midpoint and cumulative frequency. *[7]*
(b) Calculate the Mean, Median and Mode (show the median/modal class and all working). *[6]*
(c) Calculate Q_1 and Q_3 from your table. *[4]*
(d) Construct a box plot using the five-number summary (use the actual minimum and maximum). *[4]*
(e) Comment on the shape of the distribution (symmetric, positively or negatively skewed). Justify in 2–3 sentences. *[4]*

Exam Practice 3. Packet sizes (bytes) of 48 network packets (48 values):

111 123 125 126 129 133 134 134 138 144 146 147 148 149 150
151 152 152 153 154 155 156 157 157 159 159 160 161 164 165
165 168 169 170 172 172 173 173 175 175 179 180 182 183 186
190 193 200

(a) Construct a grouped frequency distribution table using a class width of 20 starting from the class 101–120. Show class limits, class boundaries, tally/frequency, midpoint and cumulative frequency. *[7]*
(b) Calculate the Mean, Median and Mode (show the median/modal class and all working). *[6]*
(c) Calculate Q_1 and Q_3 from your table. *[4]*
(d) Construct a box plot using the five-number summary (use the actual minimum and maximum). *[4]*
(e) Comment on the shape of the distribution (symmetric, positively or negatively skewed). Justify in 2–3 sentences. *[4]*

Exam Practice 4. Download speeds (Mbps) recorded on 40 devices (40 values):

```
11  12  12  13  14  14  14  14  15  15  15  15  16  16  18  18  18  19
19  20  22  22  23  23  24  24  25  26  29  30  31  32  35  35  36  38
40  42  44  47
```

(a) Construct a grouped frequency distribution table using a class width of 10 starting from the class 11–20. Show class limits, class boundaries, tally/frequency, midpoint and cumulative frequency. *[7]*
(b) Calculate the Mean, Median and Mode (show the median/modal class and all working). *[6]*
(c) Calculate Q_1 and Q_3 from your table. *[4]*
(d) Construct a box plot using the five-number summary (use the actual minimum and maximum). *[4]*
(e) Comment on the shape of the distribution (symmetric, positively or negatively skewed). Justify in 2–3 sentences. *[4]*

Exam Practice 5. Battery life (hours) of 50 laptops (50 values):

```
25  28  34  36  37  38  39  39  39  39  39  40  40  41  41  41  41  42
42  42  43  43  43  44  44  44  45  45  45  45  45  46  46  46  46  46
46  46  46  47  47  47  47  47  48  48  48  48  49  49
```

(a) Construct a grouped frequency distribution table using a class width of 5 starting from the class 21–25. Show class limits, class boundaries, tally/frequency, midpoint and cumulative frequency. *[7]*
(b) Calculate the Mean, Median and Mode (show the median/modal class and all working). *[6]*
(c) Calculate Q_1 and Q_3 from your table. *[4]*
(d) Construct a box plot using the five-number summary (use the actual minimum and maximum). *[4]*
(e) Comment on the shape of the distribution (symmetric, positively or negatively skewed). Justify in 2–3 sentences. *[4]*

Exam Practice 6. Daily failed-login attempts on a server over 45 days (45 values):

```
3   3   4   5   5   5   6   6   6   7   7   8   8   8   9   9   9   9   10   11   11   11
13  14  15  16  16  17  17  17  18  18  19  19  20  20  22  22  23  23
23  23  26  26  39
```

(a) Construct a grouped frequency distribution table using a class width of 10 starting from the class 1–10. Show class limits, class boundaries, tally/frequency, midpoint and cumulative frequency. *[7]*
(b) Calculate the Mean, Median and Mode (show the median/modal class and all working). *[6]*
(c) Calculate Q_1 and Q_3 from your table. *[4]*
(d) Construct a box plot using the five-number summary (use the actual minimum and maximum). *[4]*
(e) Comment on the shape of the distribution (symmetric, positively or negatively skewed). Justify in 2–3 sentences. *[4]*

Exam Practice 7. Model training times (minutes) for 40 runs (40 values):

```
48  50  51  51  51  55  58  60  60  61  62  62  63  63  65  65  65  66
66  66  67  68  68  68  70  72  73  74  74  75  75  76  76  76  77  77
81  81  81  86
```

(a) Construct a grouped frequency distribution table using a class width of 10 starting from the class 41–50. Show class limits, class boundaries, tally/frequency, midpoint and cumulative frequency. *[7]*
(b) Calculate the Mean, Median and Mode (show the median/modal class and all working). *[6]*
(c) Calculate Q_1 and Q_3 from your table. *[4]*
(d) Construct a box plot using the five-number summary (use the actual minimum and maximum). *[4]*
(e) Comment on the shape of the distribution (symmetric, positively or negatively skewed). Justify in 2–3 sentences. *[4]*

Exam Practice 8. Weights (g) of 50 packets of biscuits (50 values):

```
183  186  188  189  189  189  190  190  190  191  191  192  193  193  193
193  193  193  194  194  194  194  195  195  195  195  195  195  196  196
196  196  197  197  197  198  198  198  198  198  199  199  200  200  200
201  201  203  206  206
```

(a) Construct a grouped frequency distribution table using a class width of 5 starting from the class 181–185. Show class limits, class boundaries, tally/frequency, midpoint and cumulative frequency. *[7]*
(b) Calculate the Mean, Median and Mode (show the median/modal class and all working). *[6]*
(c) Calculate Q_1 and Q_3 from your table. *[4]*
(d) Construct a box plot using the five-number summary (use the actual minimum and maximum). *[4]*
(e) Comment on the shape of the distribution (symmetric, positively or negatively skewed). Justify in 2–3 sentences. *[4]*

Exam Practice 9. CPU temperatures (°C) of 48 servers (48 values):

```
47  59  64  67  69  70  73  74  74  76  77  78  78  80  80  81  81  81
81  82  82  86  86  86  86  87  87  88  89  89  89  90  90  91  91  92
93  93  93  95  95  95  95  96  96  96  96  99
```

(a) Construct a grouped frequency distribution table using a class width of 10 starting from the class 41–50. Show class limits, class boundaries, tally/frequency, midpoint and cumulative frequency. *[7]*
(b) Calculate the Mean, Median and Mode (show the median/modal class and all working). *[6]*
(c) Calculate Q_1 and Q_3 from your table. *[4]*
(d) Construct a box plot using the five-number summary (use the actual minimum and maximum). *[4]*
(e) Comment on the shape of the distribution (symmetric, positively or negatively skewed). Justify in 2–3 sentences. *[4]*

Exam Practice 10. Page load times (ms) for 50 requests (50 values):

```
31  34  35  36  36  38  39  40  41  42  42  42  42  42  43  45  45  46
46  47  48  50  51  53  53  54  55  55  56  58  60  62  64  64  65  65
67  70  70  70  75  75  77  81  81  82  88  91  92  99
```

(a) Construct a grouped frequency distribution table using a class width of 20 starting from the class 31–50. Show class limits, class boundaries, tally/frequency, midpoint and cumulative frequency. *[7]*
(b) Calculate the Mean, Median and Mode (show the median/modal class and all working). *[6]*
(c) Calculate Q_1 and Q_3 from your table. *[4]*
(d) Construct a box plot using the five-number summary (use the actual minimum and maximum). *[4]*
(e) Comment on the shape of the distribution (symmetric, positively or negatively skewed). Justify in 2–3 sentences. *[4]*

Worked Answers

Set 1: Arithmetic Mean

1.1 Find the mean of: 12, 15, 18, 20, 25.

$\Sigma X = 12 + 15 + 18 + 20 + 25 = 90$
$n = 5$
$\bar{X} = \Sigma X / n = 90 / 5 =$ **18**

1.2 Find the mean of: 45, 52, 38, 61, 49, 55.

$\Sigma X = 45 + 52 + 38 + 61 + 49 + 55 = 300$
$n = 6$
$\bar{X} = \Sigma X / n = 300 / 6 =$ **50**

1.3 Daily temperatures (°C) for a week are: 28, 31, 30, 29, 32, 33, 27.

$\Sigma X = 28 + 31 + 30 + 29 + 32 + 33 + 27 = 210$
$n = 7$
$\bar{X} = \Sigma X / n = 210 / 7 =$ **30**

1.4 Find the mean: X = 5, 10, 15, 20 with f = 3, 7, 6, 4.

X	f	fX
5	3	15
10	7	70
15	6	90
20	4	80
Total	**Σf = 20**	**ΣfX = 255**

$\bar{X} = \Sigma fX / \Sigma f = 255 / 20 =$ **12.75**

1.5 Number of children per family: X = 0, 1, 2, 3, 4 with f = 4, 8, 10, 5, 3. Find the mean.

X	f	fX
0	4	0
1	8	8
2	10	20
3	5	15
4	3	12
Total	**Σf = 30**	**ΣfX = 55**

$\bar{X} = \Sigma fX / \Sigma f = 55 / 30 =$ **1.83**

1.6 (Slide Q2) Find the mean: Class 1–10, 11–20, 21–30, 31–40; f = 3, 5, 7, 3.

Class	f	x	fx
1 – 10	3	5.5	16.5
11 – 20	5	15.5	77.5
21 – 30	7	25.5	178.5
31 – 40	3	35.5	106.5
Total	**N = 18**		**Σfx = 379**

$\bar{X}$ = Σfx / N = 379 / 18 = **21.06**

1.7 Find the mean: Class 0–10, 10–20, 20–30, 30–40, 40–50; f = 5, 8, 15, 10, 2.

Class	f	x	fx
0 – 10	5	5	25
10 – 20	8	15	120
20 – 30	15	25	375
30 – 40	10	35	350
40 – 50	2	45	90
Total	**N = 40**		**Σfx = 960**

$\bar{X}$ = Σfx / N = 960 / 40 = **24**

1.8 Find the mean: Class 21–30, 31–40, 41–50, 51–60, 61–70; f = 6, 10, 14, 7, 3.

Class	f	x	fx
21 – 30	6	25.5	153
31 – 40	10	35.5	355
41 – 50	14	45.5	637
51 – 60	7	55.5	388.5
61 – 70	3	65.5	196.5
Total	**N = 40**		**Σfx = 1730**

$\bar{X}$ = Σfx / N = 1730 / 40 = **43.25**

1.9 The mean of 6 numbers is 15. Five of them are 10, 12, 14, 18, 20. Find the sixth number.

Total of 6 numbers = n × $\bar{X}$ = 6 × 15 = 90
Sum of the five = 10 + 12 + 14 + 18 + 20 = 74
Sixth number = 90 − 74 = **16**

1.10 Section A (30 students) has mean 60; Section B (20 students) has mean 70. Find the combined mean.

Total marks A = 30 × 60 = 1800; B = 20 × 70 = 1400
Combined mean = (1800 + 1400) / (30 + 20) = 3200 / 50 = **64**
(This is a weighted mean, weights = number of students.)

Set 2: Weighted Mean

2.1 (Slide Q4) Scores 85, 90, 78 in tests worth 2, 3, 1 credit hours. Find the weighted mean.

Marks	Credit hrs	W×X
85	2	170
90	3	270
78	1	78
Total	**ΣW = 6**	**ΣWX = 518**

$\bar{X}_w = \Sigma WX / \Sigma W = 518 / 6 =$ **86.33**

2.2 Marks 75, 82, 68, 90 in subjects of 3, 4, 2, 3 credit hours. Find the weighted mean.

Marks	Credit hrs	W×X
75	3	225
82	4	328
68	2	136
90	3	270
Total	**ΣW = 12**	**ΣWX = 959**

$\bar{X}_w = \Sigma WX / \Sigma W = 959 / 12 =$ **79.92**

2.3 A shop sells rice at Rs 50/kg (10 kg), Rs 80/kg (5 kg), Rs 120/kg (2 kg). Find the average price per kg.

Price (Rs)	Qty (kg)	W×X
50	10	500
80	5	400
120	2	240
Total	**ΣW = 17**	**ΣWX = 1140**

$\bar{X}_w = \Sigma WX / \Sigma W = 1140 / 17 =$ **67.06**

2.4 Grade points 4.0, 3.3, 2.7, 3.7 for courses of 3, 3, 2, 4 credit hours. Find the GPA.

Grade pts	Credit hrs	W×X
4.0	3	12
3.3	3	9.9
2.7	2	5.4
3.7	4	14.8
Total	**ΣW = 12**	**ΣWX = 42.1**

$\bar{X}_w = \Sigma WX / \Sigma W = 42.1 / 12 =$ **3.51**

2.5 Quiz = 20%, Mid = 30%, Final = 50%. Scores: 80, 70, 85. Find the overall score.

Score	Weight %	W×X
80	20	1600
70	30	2100
85	50	4250
Total	**ΣW = 100**	**ΣWX = 7950**

$\bar{X}_w = \Sigma WX / \Sigma W = 7950 / 100 =$ **79.5**

2.6 Values 60, 70, 80 all have weight 2. Find the weighted mean and compare with the simple mean.

X	W	W×X
60	2	120
70	2	140
80	2	160
Total	**ΣW = 6**	**ΣWX = 420**

$\bar{X}_w = \Sigma WX / \Sigma W = 420 / 6 =$ **70**
All weights equal → weighted mean = simple mean (60+70+80)/3 = 70 ✓

2.7 Marks 80 (weight 2) and 60 (weight w) give a weighted mean of 70. Find w.

(80×2 + 60w) / (2 + w) = 70
160 + 60w = 140 + 70w
20 = 10w → **w = 2**

2.8 Values 90 (weight 1) and 60 (weight 4). Find the simple mean and the weighted mean.

Simple mean = (90 + 60)/2 = 75

X	W	W×X
90	1	90
60	4	240
Total	**ΣW = 5**	**ΣWX = 330**

$\bar{X}_w = \Sigma WX / \Sigma W = 330 / 5 =$ **66**
Weighted mean is pulled toward 60 because it has the larger weight.

2.9 (AI) A model scores 92% accuracy on 1000 test samples and 85% on 4000 samples. Find the overall accuracy.

Accuracy %	Samples	W×X
92	1000	92000
85	4000	340000
Total	**ΣW = 5000**	**ΣWX = 432000**

$\bar{X}_w = \Sigma WX / \Sigma W = 432000 / 5000 =$ **86.4**

2.10 (Cyber) Server A: average response 100 ms over 200 requests; Server B: 150 ms over 300 requests. Find the overall average response time.

Avg time (ms)	Requests	W×X
100	200	20000
150	300	45000
Total	**ΣW = 500**	**ΣWX = 65000**

$\bar{X}_w = \Sigma WX / \Sigma W = 65000 / 500 =$ **130**

Set 3: Median

3.1 Find the median of: 7, 3, 9, 1, 5.

Sorted: 1, 3, 5, 7, 9
n = 5 (odd) → middle (3th) value = **5**

3.2 Find the median of: 12, 5, 8, 20, 15, 10.

Sorted: 5, 8, 10, 12, 15, 20
n = 6 (even) → (10 + 12) / 2 = **11**

3.3 Find the median of: 23, 19, 31, 27, 19, 25, 30.

Sorted: 19, 19, 23, 25, 27, 30, 31
n = 7 (odd) → middle (4th) value = **25**

3.4 Response times (ms): 100, 105, 98, 250, 102, 101. Find the median. Why is it better than the mean here?

Sorted: 98, 100, 101, 102, 105, 250
n = 6 (even) → (101 + 102) / 2 = **101.5**
Mean = 756/6 = 126 → the outlier 250 pulls the mean up; the median (101.5) better represents a typical day.

3.5 Find the median: X = 1, 2, 3, 4, 5; f = 3, 5, 8, 4, 2.

X	f	c.f.
1	3	3
2	5	8
3	8	16
4	4	20
5	2	22

n = 22 → n/2 = 11
First c.f. ≥ 11 is 16 (at X = 3) → Median = **3**

3.6 Find the median: X = 10, 20, 30, 40, 50; f = 4, 6, 5, 3, 2.

X	f	c.f.
10	4	4
20	6	10
30	5	15
40	3	18
50	2	20

n = 20 → n/2 = 10
First c.f. ≥ 10 is 10 (at X = 20) → Median = **20**

3.7 (Slide Q2) Find the median: Class 1–10, 11–20, 21–30, 31–40; f = 3, 5, 7, 3.

Class	Boundary	f	c.f.
1 – 10	0.5 – 10.5	3	3
11 – 20	10.5 – 20.5	5	8
21 – 30	20.5 – 30.5	7	15
31 – 40	30.5 – 40.5	3	18
Total		**N = 18**	

Median: position = N/2 = 9 → class 21 – 30 (boundary 20.5 – 30.5)
L = 20.5, cf = 8, f = 7, c = 10
Median = 20.5 + [(9 − 8) / 7] × 10 = 20.5 + 1.43 = **21.93**

3.8 Find the median: Class 0–10, 10–20, 20–30, 30–40, 40–50; f = 5, 8, 15, 10, 2. (No gaps, so boundaries = limits.)

Class	Boundary	f	c.f.
0 – 10	0 – 10	5	5
10 – 20	10 – 20	8	13
20 – 30	20 – 30	15	28
30 – 40	30 – 40	10	38
40 – 50	40 – 50	2	40
Total		**N = 40**	

Median: position = N/2 = 20 → class 20 – 30 (boundary 20 – 30)
L = 20, cf = 13, f = 15, c = 10
Median = 20 + [(20 − 13) / 15] × 10 = 20 + 4.67 = **24.67**

3.9 Find the median: Class 21–30, 31–40, 41–50, 51–60, 61–70; f = 6, 10, 14, 7, 3.

Class	Boundary	f	c.f.
21 – 30	20.5 – 30.5	6	6
31 – 40	30.5 – 40.5	10	16
41 – 50	40.5 – 50.5	14	30
51 – 60	50.5 – 60.5	7	37
61 – 70	60.5 – 70.5	3	40
Total		**N = 40**	

Median: position = N/2 = 20 → class 41 – 50 (boundary 40.5 – 50.5)
L = 40.5, cf = 16, f = 14, c = 10
Median = 40.5 + [(20 − 16) / 14] × 10 = 40.5 + 2.86 = **43.36**

3.10 Find the median: Class 100–119, 120–139, 140–159, 160–179, 180–199; f = 8, 14, 20, 12, 6.

Class	Boundary	f	c.f.
100 – 119	99.5 – 119.5	8	8
120 – 139	119.5 – 139.5	14	22
140 – 159	139.5 – 159.5	20	42
160 – 179	159.5 – 179.5	12	54
180 – 199	179.5 – 199.5	6	60
Total		**N = 60**	

Median: position = N/2 = 30 → class 140 – 159 (boundary 139.5 – 159.5)
L = 139.5, cf = 22, f = 20, c = 20
Median = 139.5 + [(30 − 22) / 20] × 20 = 139.5 + 8 = **147.5**

Set 4: Mode and the Empirical Relationship

4.1 Find the mode: 2, 3, 3, 5, 7, 3, 8.

3 appears 3 times (most often) → **Mode = 3** (unimodal)

4.2 Find the mode: 4, 6, 6, 8, 9, 9, 10.

6 and 9 each appear twice → **Modes = 6 and 9** (bimodal)

4.3 Find the mode: 1, 2, 3, 4, 5.

Every value appears once → **No mode**

4.4 Shoe sizes sold: size 6, 7, 8, 9, 10 with f = 5, 12, 20, 15, 8. Find the mode.

Highest frequency = 20 at size 8 → **Mode = 8**

4.5 (Slide Q2) Find the mode: Class 1–10, 11–20, 21–30, 31–40; f = 3, 5, 7, 3.

Class	Boundary	f
1 – 10	0.5 – 10.5	3
11 – 20	10.5 – 20.5	5
21 – 30	20.5 – 30.5	7
31 – 40	30.5 – 40.5	3
Total		**N = 18**

Modal class = 21 – 30 (highest f = 7); boundary 20.5 – 30.5
L = 20.5, f_1 = 7, f_o = 5, f_2 = 3, c = 10
Mode = 20.5 + [(7 − 5) / (2×7 − 5 − 3)] × 10 = 20.5 + [2/6] × 10 = **23.83**

4.6 Find the mode: Class 0–10, 10–20, 20–30, 30–40, 40–50; f = 5, 8, 15, 10, 2.

Class	Boundary	f
0 – 10	0 – 10	5
10 – 20	10 – 20	8
20 – 30	20 – 30	15
30 – 40	30 – 40	10
40 – 50	40 – 50	2
Total		**N = 40**

Modal class = 20 – 30 (highest f = 15); boundary 20 – 30
L = 20, f_1 = 15, f_o = 8, f_2 = 10, c = 10
Mode = 20 + [(15 − 8) / (2×15 − 8 − 10)] × 10 = 20 + [7/12] × 10 = **25.83**

4.7 Find the mode: Class 21–30, 31–40, 41–50, 51–60, 61–70; f = 6, 10, 14, 7, 3.

Class	Boundary	f
21 – 30	20.5 – 30.5	6
31 – 40	30.5 – 40.5	10
41 – 50	40.5 – 50.5	14
51 – 60	50.5 – 60.5	7
61 – 70	60.5 – 70.5	3
Total		**N = 40**

Modal class = 41 – 50 (highest f = 14); boundary 40.5 – 50.5
L = 40.5, f_1 = 14, f_o = 10, f_2 = 7, c = 10
Mode = 40.5 + [(14 − 10) / (2×14 − 10 − 7)] × 10 = 40.5 + [4/11] × 10 = **44.14**

4.8 Find the mode: Class 0–10, 10–20, 20–30, 30–40; f = 12, 8, 5, 3.

Class	Boundary	f
0 – 10	0 – 10	12
10 – 20	10 – 20	8
20 – 30	20 – 30	5
30 – 40	30 – 40	3
Total		**N = 28**

Modal class = 0 – 10 (highest f = 12); boundary 0 – 10
L = 0, f_1 = 12, f_o = 0, f_2 = 8, c = 10
Mode = 0 + [(12 − 0) / (2×12 − 0 − 8)] × 10 = 0 + [12/16] × 10 = **7.5**
Note: the modal class is the FIRST class, so there is no class before it → f_o = 0.

4.9 Mean = 50, Median = 48. Estimate the mode.

Mode = 3 × Median − 2 × Mean = 3(48) − 2(50) = 144 − 100 = **44**

4.10 Mode = 30, Mean = 36. Estimate the median.

Mode = 3 Median − 2 Mean → 30 = 3 Median − 72
3 Median = 102 → **Median = 34**

Set 5: Quartiles, IQR, QD and Box Plot

5.1 Find Q_1, Q_2, Q_3, IQR and QD: 3, 7, 8, 12, 13, 14, 18, 21.

Sorted: 3, 7, 8, 12, 13, 14, 18, 21 (n = 8)
Q_1: position 1(n+1)/4 = 2.25 → between 2th (7) and 3th (8) → 7 + 0.25(8 − 7) = **7.25**
Q_2: position 2(n+1)/4 = 4.5 → between 4th (12) and 5th (13) → 12 + 0.5(13 − 12) = **12.5**
Q_3: position 3(n+1)/4 = 6.75 → between 6th (14) and 7th (18) → 14 + 0.75(18 − 14) = **17**
IQR = 17 − 7.25 = **9.75**; QD = **4.88**

5.2 Find Q_1, Q_2, Q_3: 2, 4, 6, 8, 10, 12, 14.

Sorted: 2, 4, 6, 8, 10, 12, 14 (n = 7)
Q_1: position 1(n+1)/4 = 2 → 2th value = **4**
Q_2: position 2(n+1)/4 = 4 → 4th value = **8**
Q_3: position 3(n+1)/4 = 6 → 6th value = **12**

5.3 Find Q_1, Q_2, Q_3 and IQR: 15, 9, 22, 11, 18, 25, 13, 20, 17, 10, 14.

Sorted: 9, 10, 11, 13, 14, 15, 17, 18, 20, 22, 25 (n = 11)
Q_1: position 1(n+1)/4 = 3 → 3th value = **11**
Q_2: position 2(n+1)/4 = 6 → 6th value = **15**
Q_3: position 3(n+1)/4 = 9 → 9th value = **20**
IQR = **9**

5.4 (Slide Q3) Find Q_1, Q_3, IQR and QD: Class 1–10, 11–20, 21–30, 31–40; f = 3, 5, 7, 3.

Class	Boundary	f	c.f.
1 – 10	0.5 – 10.5	3	3
11 – 20	10.5 – 20.5	5	8
21 – 30	20.5 – 30.5	7	15
31 – 40	30.5 – 40.5	3	18
Total		**N = 18**	

Q_1: position = N/4 = 4.5 → class 11 – 20 (boundary 10.5 – 20.5)
L = 10.5, cf = 3, f = 5, c = 10
$Q_1 = 10.5 + [(4.5 - 3) / 5] \times 10 = 10.5 + 3 =$ **13.5**
Q_3: position = 3N/4 = 13.5 → class 21 – 30 (boundary 20.5 – 30.5)
L = 20.5, cf = 8, f = 7, c = 10
$Q_3 = 20.5 + [(13.5 - 8) / 7] \times 10 = 20.5 + 7.86 =$ **28.36**
IQR = 28.36 − 13.5 = **14.86**; QD = **7.43**

5.5 Find Q_1, Q_3 and IQR: Class 0–10, 10–20, 20–30, 30–40, 40–50; f = 5, 8, 15, 10, 2.

Class	f	c.f.
0 – 10	5	5
10 – 20	8	13
20 – 30	15	28
30 – 40	10	38
40 – 50	2	40
Total	**N = 40**	

Q_1: position = N/4 = 10 → class 10 – 20 (boundary 10 – 20)
L = 10, cf = 5, f = 8, c = 10
$Q_1 = 10 + [(10 - 5) / 8] \times 10 = 10 + 6.25 =$ **16.25**
Q_3: position = 3N/4 = 30 → class 30 – 40 (boundary 30 – 40)
L = 30, cf = 28, f = 10, c = 10
$Q_3 = 30 + [(30 - 28) / 10] \times 10 = 30 + 2 =$ **32**
IQR = **15.75**

5.6 Find Q_1, Q_3, IQR and QD: Class 21–30, 31–40, 41–50, 51–60, 61–70; f = 6, 10, 14, 7, 3.

Class	Boundary	f	c.f.
21 – 30	20.5 – 30.5	6	6
31 – 40	30.5 – 40.5	10	16
41 – 50	40.5 – 50.5	14	30
51 – 60	50.5 – 60.5	7	37
61 – 70	60.5 – 70.5	3	40
Total		**N = 40**	

Q_1: position = N/4 = 10 → class 31 – 40 (boundary 30.5 – 40.5)
L = 30.5, cf = 6, f = 10, c = 10
$Q_1 = 30.5 + [(10 - 6) / 10] \times 10 = 30.5 + 4 =$ **34.5**
Q_3: position = 3N/4 = 30 → class 41 – 50 (boundary 40.5 – 50.5)
L = 40.5, cf = 16, f = 14, c = 10
$Q_3 = 40.5 + [(30 - 16) / 14] \times 10 = 40.5 + 10 =$ **50.5**
IQR = **16**; QD = **8**

5.7 $Q_1 = 40$ and $Q_3 = 64$. Find the IQR and the Quartile Deviation.

IQR = $Q_3 - Q_1 = 64 - 40 =$ **24**
QD = IQR / 2 = **12**

5.8 Give the five-number summary for 3, 7, 8, 12, 13, 14, 18, 21 and describe its box plot.

Min = 3, $Q_1 = 7.25$, Median = 12.5, $Q_3 = 17$, Max = 21
Box from 7.25 to 17 with a line at 12.5; whiskers to 3 and 21.

5.9 Section A: $Q_1 = 50$, $Q_3 = 70$. Section B: $Q_1 = 45$, $Q_3 = 80$. Which section's middle 50% is more spread out?

IQR A = 20, IQR B = 35 → **Section B** is more spread out

5.10 For 2, 4, 6, 8, 10, 12, 14 show that Q_2 equals the median.

Q_2 position = 2(7+1)/4 = 4 → 4th value = 8
Median (n = 7, odd) = 4th value = 8 → **Q_2 = Median = 8** ✓

Set 6: Range and Coefficient of Range

6.1 (Slide Q5) Find the range and coefficient of range: 12, 18, 9, 25, 30.

L = 30, S = 9
Range = 30 − 9 = **21**
Coeff. of Range = (30 − 9) / (30 + 9) = 21/39 = **0.5385** = 53.85%

6.2 Find the range and coefficient of range: 40, 55, 32, 70, 48.

L = 70, S = 32
Range = 70 − 32 = **38**
Coeff. of Range = (70 − 32) / (70 + 32) = 38/102 = **0.3725** = 37.25%

6.3 Temperatures (°C): 18, 25, 30, 22, 35. Find the range and coefficient of range.

L = 35, S = 18
Range = 35 − 18 = **17 °C**
Coeff. of Range = (35 − 18) / (35 + 18) = 17/53 = **0.3208** = 32.08%

6.4 Prices range from Rs 250 to Rs 400. Find the coefficient of range.

Coeff. = (400 − 250)/(400 + 250) = 150/650 = **0.2308** = 23.08%

6.5 Exam marks: lowest 33, highest 97. Find the range and coefficient of range.

Range = 97 − 33 = **64**
Coeff. = 64/130 = **0.4923** = 49.23%

6.6 The largest value is 95 and the coefficient of range is 0.25. Find the smallest value.

(95 − S)/(95 + S) = 0.25
95 − S = 23.75 + 0.25S
71.25 = 1.25S → **S = 57**

6.7 Response times (ms): 120, 135, 128, 410, 125. Find the range. What limitation does this show?

Range = 410 − 120 = **290 ms**
Limitation: one extreme value (410) makes the spread look huge; range ignores all values in between.

6.8 A: 10, 10, 10, 10, 50 and B: 10, 20, 30, 40, 50. Find both ranges. What does this show?

Range A = 40, Range B = 40 → **same range**
But B is spread evenly and A is not → range uses only 2 values, so it can hide differences.

6.9 (Slide 27 data) Class A: 45, 46, 47, 48, 49; Class B: 20, 35, 50, 65, 80. Compare using coefficient of range.

A: (49 − 45)/(49 + 45) = 4/94 = **0.0426**
B: (80 − 20)/(80 + 20) = 60/100 = **0.6**
→ B is far more spread out.

6.10 Heights: 150–190 cm. Weights: 45–90 kg. Which varies more? (Use coefficient of range.)

Heights: 40/340 = **0.1176**
Weights: 45/135 = **0.3333**
→ **Weights** vary more (units differ, so use the coefficient).

Set 7: Variance and Standard Deviation

7.1 (Slide Q6) Find the variance and S.D.: 3, 6, 9, 12, 15.

$\bar{X} = 45 / 5 = 9$

X	X − $\bar{X}$	(X − $\bar{X}$)²
3	-6	36
6	-3	9
9	0	0
12	3	9
15	6	36
Total	**0**	**90**

$\sigma^2 = \Sigma(X - \bar{X})^2 / n = 90 / 5 =$ **18**
$\sigma = \sqrt{18} =$ **4.24**

7.2 Find the variance and S.D.: 10, 20, 30, 40, 50.

$\bar{X} = 150 / 5 = 30$

X	X − $\bar{X}$	(X − $\bar{X}$)²
10	-20	400
20	-10	100
30	0	0
40	10	100
50	20	400
Total	**0**	**1000**

$\sigma^2 = \Sigma(X - \bar{X})^2 / n = 1000 / 5 =$ **200**
$\sigma = \sqrt{200} =$ **14.14**

7.3 Find the variance and S.D.: 5, 7, 9, 11, 13, 15.

$\bar{X} = 60 / 6 = 10$

X	X − $\bar{X}$	(X − $\bar{X}$)²
5	-5	25
7	-3	9
9	-1	1
11	1	1
13	3	9
15	5	25
Total	**0**	**70**

$\sigma^2 = \Sigma(X - \bar{X})^2 / n = 70 / 6 =$ **11.67**
$\sigma = \sqrt{11.67} =$ **3.42**

7.4 (Slide 27, Class A) Find the variance and S.D.: 45, 46, 47, 48, 49.

$\bar{X} = 235 / 5 = 47$

X	X − $\bar{X}$	(X − $\bar{X}$)²
45	-2	4
46	-1	1
47	0	0
48	1	1
49	2	4
Total	**0**	**10**

$\sigma^2 = \Sigma(X - \bar{X})^2 / n = 10 / 5 =$ **2**
$\sigma = \sqrt{2} =$ **1.41**

7.5 (Slide 27, Class B) Find the variance and S.D.: 20, 35, 50, 65, 80.

$\bar{X} = 250 / 5 = 50$

X	X − $\bar{X}$	(X − $\bar{X}$)²
20	-30	900
35	-15	225
50	0	0
65	15	225
80	30	900
Total	**0**	**2250**

$\sigma^2 = \Sigma(X - \bar{X})^2 / n = 2250 / 5 =$ **450**
$\sigma = \sqrt{450} =$ **21.21**

7.6 Find the variance and S.D.: X = 1, 2, 3, 4, 5; f = 4, 8, 10, 5, 3.

X	f	fX	fX^2
1	4	4	4
2	8	16	32
3	10	30	90
4	5	20	80
5	3	15	75
Total	**30**	**85**	**281**

$\bar{X} = 85/30 = 2.83$
$\sigma^2 = \Sigma fX^2/N - (\Sigma fX/N)^2 = 281/30 - (2.83)^2 = 9.37 - 8.03 =$ **1.34**
$\sigma =$ **1.16**

7.7 Find the variance and S.D.: Class 1–10, 11–20, 21–30, 31–40; f = 3, 5, 7, 3.

Class	f	x	fx	fx^2
1 – 10	3	5.5	16.5	90.75
11 – 20	5	15.5	77.5	1201.25
21 – 30	7	25.5	178.5	4551.75
31 – 40	3	35.5	106.5	3780.75
Total	**N = 18**		**Σfx = 379**	**Σfx^2 = 9624.5**

$\sigma^2 = \Sigma fx^2/N - (\Sigma fx/N)^2 = 9624.5/18 - (379/18)^2 = 534.69 - (21.06)^2 = 534.69 - 443.34 =$ **91.36**
$\sigma = \sqrt{91.36} =$ **9.56**

7.8 Find the variance and S.D.: Class 0–10, 10–20, 20–30, 30–40, 40–50; f = 5, 8, 15, 10, 2.

Class	f	x	fx	fx^2
0 – 10	5	5	25	125
10 – 20	8	15	120	1800
20 – 30	15	25	375	9375
30 – 40	10	35	350	12250
40 – 50	2	45	90	4050
Total	**N = 40**		**Σfx = 960**	**Σfx^2 = 27600**

$\sigma^2 = \Sigma fx^2/N - (\Sigma fx/N)^2 = 27600/40 - (960/40)^2 = 690 - (24)^2 = 690 - 576 =$ **114**
$\sigma = \sqrt{114} =$ **10.68**

7.9 Find the variance and S.D.: Class 21–30, 31–40, 41–50, 51–60, 61–70; f = 6, 10, 14, 7, 3.

Class	f	x	fx	fx^2
21 – 30	6	25.5	153	3901.5
31 – 40	10	35.5	355	12602.5
41 – 50	14	45.5	637	28983.5
51 – 60	7	55.5	388.5	21561.75
61 – 70	3	65.5	196.5	12870.75
Total	**N = 40**		**Σfx = 1730**	**Σfx^2 = 79920**

$\sigma^2 = \Sigma fx^2/N - (\Sigma fx/N)^2 = 79920/40 - (1730/40)^2 = 1998 - (43.25)^2 = 1998 - 1870.56 =$ **127.44**
$\sigma = \sqrt{127.44} =$ **11.29**

7.10 For 4, 8, 6, 2, 10 find the population variance (÷n) and the sample variance (÷(n − 1)).

$\bar{X} = 30 / 5 = 6$

X	X − $\bar{X}$	(X − $\bar{X}$)²
4	-2	4
8	2	4
6	0	0
2	-4	16
10	4	16
Total	**0**	**40**

$\sigma^2 = \Sigma(X - \bar{X})^2 / n = 40 / 5 =$ **8**
$\sigma = \sqrt{8} =$ **2.83**
Sample variance (Excel VAR.S) = $\Sigma(X - \bar{X})^2/(n - 1) = 40/4 =$ **10**

Set 8: Coefficient of Variation

8.1 (Slide Q7) Machines A and B: mean 10 mm; S.D. A = 0.5 mm, B = 0.8 mm. Which is more consistent?

C.V.(A) = 0.5/10 × 100 = **5%**
C.V.(B) = 0.8/10 × 100 = **8%**
→ **Machine A** is more consistent (lower C.V.)

8.2 X: mean 40, S.D. 8. Y: mean 120, S.D. 18. Which is more consistent?

C.V.(X) = 8/40 × 100 = **20%**
C.V.(Y) = 18/120 × 100 = **15%**
→ **Y** is more consistent, even though its S.D. is larger

8.3 Heights: mean 160 cm, S.D. 8. Weights: mean 60 kg, S.D. 6. Which varies more?

C.V. heights = 5%, C.V. weights = **10%** → **weights** vary more

8.4 Batsman A scores 45, 50, 55, 60, 40; Batsman B scores 20, 90, 35, 80, 25. Who is more consistent?

A: mean = 50, σ = 7.07, C.V. = **14.14%**
B: mean = 50, σ = 29.15, C.V. = **58.31%**
→ **Batsman A** is more consistent

8.5 (Slide Example 4.9) Prices: 8, 13, 18, 23, 30. Life (hrs): 130, 150, 180, 250, 345. Which has greater relative variation?

Price: mean 18.4, σ 7.66, C.V. = **41.62%**
Life: mean 211, σ 78.38, C.V. = **37.15%**
→ **Price** varies more relatively (despite the smaller S.D.)

8.6 Using your answers for Class 1–10 … 31–40 (f = 3, 5, 7, 3), find the C.V.

Mean = 21.06, σ = 9.56
C.V. = 9.56/21.06 × 100 = **45.39%**

8.7 C.V. = 25% and mean = 40. Find the S.D.

σ = C.V. × mean / 100 = 25 × 40 / 100 = **10**

8.8 C.V. = 12% and S.D. = 3. Find the mean.

Mean = σ × 100 / C.V. = 3 × 100 / 12 = **25**

8.9 (AI) Model A: mean accuracy 90%, S.D. 2. Model B: mean 85%, S.D. 1.5 (across folds). Which is more stable?

C.V.(A) = 2/90 × 100 = **2.22%**
C.V.(B) = 1.5/85 × 100 = **1.76%**
→ **Model B** is more stable

8.10 (Slide 27 data) Class A: mean 47, σ = 1.41; Class B: mean 50, σ = 21.21. Compare consistency.

C.V.(A) = 1.41/47 × 100 = **3%**
C.V.(B) = 21.21/50 × 100 = **42.42%**
→ **Class A** is far more consistent

Set 9: Skewness

9.1 (Slide Q9) Mean = 32, Median = 30, Mode = 26, S.D. = 8. Find Sk (using mode). Type?

Sk = (32 − 26)/8 = **0.75** → **positively skewed**

9.2 Mean = 45, Median = 48, Mode = 50, S.D. = 10. Find Sk by both formulas.

Sk = (45 − 50)/10 = **−0.5**
Sk = 3(45 − 48)/10 = **−0.9**
→ **negatively skewed**

9.3 Mean = Median = Mode = 40, S.D. = 5. Find Sk.

Sk = (40 − 40)/5 = **0** → **symmetric**

9.4 Mean = 60, Median = 55, S.D. = 15 (mode not given). Find Sk.

Use the median formula: Sk = 3(60 − 55)/15 = 15/15 = **1.0** → **positively skewed**

9.5 (Slide example) Mean = 24.83, Median = 25.5, Mode = 27.17, S.D. = 11.24. Find Sk by both formulas.

Sk = (24.83 − 27.17)/11.24 = **−0.21**
Sk = 3(24.83 − 25.5)/11.24 = **−0.18**
→ slightly **negatively skewed**

9.6 Mean = 70, Median = 66, S.D. = 12. Estimate the mode, then find Sk.

Mode = 3(66) − 2(70) = 198 − 140 = 58
Sk = (70 − 58)/12 = **1.0** (and 3(70 − 66)/12 = 1.0 ✓) → positive

9.7 Sk = 0.5, Mean = 40, S.D. = 8. Find the mode.

0.5 = (40 − Mode)/8 → 40 − Mode = 4 → **Mode = 36**

9.8 Mean = 50, Median = 52, Mode = 55. Without calculating, state the skewness.

Mean < Median < Mode → **negatively (left) skewed**

9.9 Data: 2, 3, 3, 3, 4, 5, 9. Find the mean, median, mode, S.D. and Sk (using mode).

Mean = 29/7 = 4.14, Median = 3, Mode = 3
σ = √(32.86/7) = 2.17
Sk = (4.14 − 3)/2.17 = **0.53** → **positively skewed** (tail toward 9)

9.10 Income data usually has a long right tail. Which is larger, the mean or the median? Why?

Mean > Median — positive skew: a few very high incomes pull the mean to the right (the mean follows the tail).

Set 10: Kurtosis

10.1 (Slide Q10) Excel reports excess kurtosis = +2.5. Type? Outliers?

Excess > 0 → **Leptokurtic** (peaked, fat tails)
→ **more extreme outliers** than a Normal distribution

10.2 Find the kurtosis of 2, 4, 6, 8, 10 using $\beta_2 = m_4/m_2^2$.

Mean = 6; deviations −4, −2, 0, 2, 4
$m_2 = (16+4+0+4+16)/5 = 8$
$m_4 = (256+16+0+16+256)/5 = 108.8$
$\beta_2 = 108.8/8^2 = 108.8/64 =$ **1.7** < 3 → **Platykurtic** (excess = -1.3)

10.3 Find the kurtosis of 1, 5, 5, 5, 5, 5, 9.

Mean = 5; deviations −4, 0, 0, 0, 0, 0, 4
$m_2 = 32/7 = 4.57$
$m_4 = 512/7 = 73.14$
$\beta_2 = 73.14/4.57^2 =$ **3.5** > 3 → **Leptokurtic** (peak at 5, extreme tails)

10.4 Excel KURT = −1.2. Type?

Excess < 0 → **Platykurtic** (flat, fewer/milder outliers)

10.5 $\beta_2 = 3$. Type? What is the excess kurtosis?

Excess = 3 − 3 = 0 → **Mesokurtic** (same as Normal)

10.6 $m_2 = 4$ and $m_4 = 40$. Find β_2 and the type.

$\beta_2 = 40/4^2 = 40/16 =$ **2.5** < 3 → **Platykurtic** (excess −0.5)

10.7 $m_2 = 5$ and $m_4 = 100$. Find β_2 and the type.

$\beta_2 = 100/5^2 = 100/25 =$ **4** > 3 → **Leptokurtic** (excess +1)

10.8 Kurtosis $\beta_2 = 4.2$. Find the excess kurtosis and the type.

Excess = 4.2 − 3 = **1.2** > 0 → **Leptokurtic**

10.9 Dataset A: excess kurtosis 1.5; Dataset B: −0.8. Which has fatter tails?

A (leptokurtic, excess > 0) has fatter tails / more outliers; B is platykurtic

10.10 (Cyber) Login-attempt data has kurtosis $\beta_2 = 7$. What does this suggest?

Excess = 4 > 0 → **Leptokurtic**
Rare-but-extreme events (possible attacks) happen more often than a Normal model predicts → monitor closely.

Set 11: Table Construction & Box Plot Basics

11.1 Class limits 31–50. Find the class boundaries, class width and midpoint.

Boundaries 30.5 – 50.5; width = 50.5 − 30.5 = **20**; midpoint = (31 + 50)/2 = **40.5**

11.2 Classes 10–19, 20–29, 30–39. Write the class boundaries.

9.5–19.5, 19.5–29.5, 29.5–39.5 (gap between classes = 1, so subtract/add 0.5)

11.3 Class boundaries 50.5 – 70.5. Find the class limits (integer data) and midpoint.

Limits **51 – 70**; midpoint = **60.5**

11.4 Data range 40 to 167, width 20, starting at 31–50. List all classes.

31–50, 51–70, 71–90, 91–110, 111–130, 131–150, **151–170** (7 classes — last must contain 167)

11.5 Make the tally and frequency for the class 41–50 from: 38, 41, 45, 50, 51, 47, 49, 44, 52, 50.

Values in 41–50: 41, 45, 50, 47, 49, 44, 50 → 𝍸 || **f = 7** (38, 51, 52 are outside)

11.6 Frequencies 6, 11, 15, 9, 4. Write the cumulative frequencies.

6, 17, 32, 41, 45 (N = 45)

11.7 N = 50 and c.f. are 17, 30, 41, 47, 48, 49, 50. Which class contains the median? Q_1? Q_3?

N/2 = 25 → **2nd** class (c.f. 30); N/4 = 12.5 → **1st** class (c.f. 17); 3N/4 = 37.5 → **3rd** class (c.f. 41)

11.8 The modal class is the first class with f = 17; the next class has f = 13. What are f_o and f_2?

f_o = 0 (no class before), **f_2 = 13**

11.9 Five-number summary: 40, 45.2, 62.8, 84.1, 167. Which whisker is longer? What does it suggest?

Left whisker = 5.2, right whisker = 82.9 → **right** whisker much longer → long right tail → **positive skew**

11.10 Mean = 70.4, Median = 62.8, Mode = 46.7. State the shape and justify.

Mean > Median > Mode → **positively (right) skewed**; a few very large values pull the mean to the right.

Exam-Style Practice — Full Solutions

Exam Practice 1 — Solution

(a) Range = 33 − 1 = 32. Classes of width 10 starting 1–10; last class 31–40 contains the maximum 33. Boundaries = lower − 0.5 and upper + 0.5; midpoint = (lower + upper)/2.

Class limits	Class boundaries	Tally	f	Midpoint x	c.f.
1 – 10	0.5 – 10.5	𝍸 𝍸 𝍸 ||||	19	5.5	19
11 – 20	10.5 – 20.5	𝍸 𝍸 𝍸 ||	17	15.5	36
21 – 30	20.5 – 30.5	𝍸 𝍸 |	11	25.5	47
31 – 40	30.5 – 40.5	|||	3	35.5	50
Total			**N = 50**		

Check: Σf = 50 = number of observations ✓

(b)

Class	f	x	fx
1 – 10	19	5.5	104.5
11 – 20	17	15.5	263.5
21 – 30	11	25.5	280.5
31 – 40	3	35.5	106.5
Total	**50**		**755**

Mean = Σfx / N = 755 / 50 = **15.1 s**

Median: N/2 = 25 → first c.f. ≥ 25 is 36 → **median class 11 – 20** (boundary 10.5 – 20.5).
L = 10.5, cf = 19, f = 17, c = 10
Median = 10.5 + [(25 − 19) / 17] × 10 = 10.5 + 3.53 = **14.03 s**

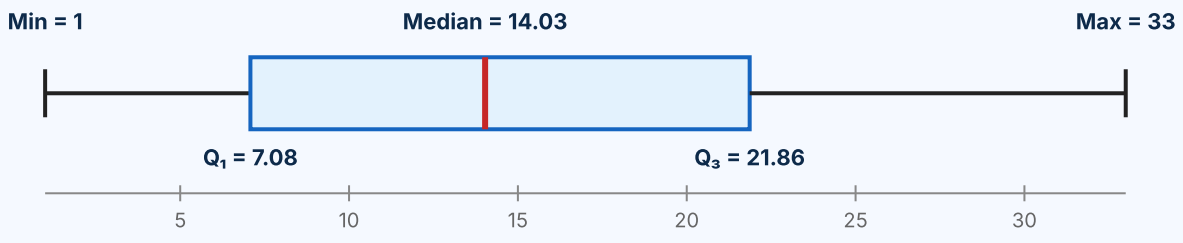
Mode: highest f = 19 → **modal class 1 – 10** (boundary 0.5 – 10.5).
L = 0.5, f_1 = 19, f_o = 0 (no class before), f_2 = 17, c = 10
Mode = 0.5 + [(19 − 0) / (2×19 − 0 − 17)] × 10 = 0.5 + 9.05 = **9.55 s**

(c)

Q_1: N/4 = 12.5 → class 1 – 10 (L = 0.5, cf = 0, f = 19, c = 10)
Q_1 = 0.5 + [(12.5 − 0) / 19] × 10 = 0.5 + 6.58 = **7.08 s**

Q_3: 3N/4 = 37.5 → class 21 – 30 (L = 20.5, cf = 36, f = 11, c = 10)
Q_3 = 20.5 + [(37.5 − 36) / 11] × 10 = 20.5 + 1.36 = **21.86 s**

(d) Five-number summary: Min = **1**, Q_1 = **7.08**, Median = **14.03**, Q_3 = **21.86**, Max = **33**. IQR = 14.78.



(e) Mean = 15.1, Median = 14.03, Mode = 9.55 → **Mean > Median > Mode**. In the box plot the right whisker is much longer than the left (left 6.08, right 11.14) and the median line lies closer to Q_1 (right half of the box is wider). Therefore the distribution is **positively (right) skewed**.

Exam Practice 2 — Solution

(a) Range = 89 − 38 = 51. Classes of width 10 starting 31–40; last class 81–90 contains the maximum 89. Boundaries = lower − 0.5 and upper + 0.5; midpoint = (lower + upper)/2.

Class limits	Class boundaries	Tally	f	Midpoint x	c.f.
31 – 40	30.5 – 40.5	|	1	35.5	1
41 – 50	40.5 – 50.5	||	2	45.5	3
51 – 60	50.5 – 60.5	|	1	55.5	4
61 – 70	60.5 – 70.5	𝍸 |	6	65.5	10
71 – 80	70.5 – 80.5	𝍸 𝍸 𝍸 |	16	75.5	26
81 – 90	80.5 – 90.5	𝍸 𝍸 𝍸 ||||	19	85.5	45
Total			**N = 45**		

Check: Σf = 45 = number of observations ✓

(b)

Class	f	x	fx
31 – 40	1	35.5	35.5
41 – 50	2	45.5	91
51 – 60	1	55.5	55.5
61 – 70	6	65.5	393
71 – 80	16	75.5	1208
81 – 90	19	85.5	1624.5
Total	**45**		**3407.5**

Mean = Σfx / N = 3407.5 / 45 = **75.72 marks**

Median: N/2 = 22.5 → first c.f. ≥ 22.5 is 26 → **median class 71 – 80** (boundary 70.5 – 80.5).
L = 70.5, cf = 10, f = 16, c = 10
Median = 70.5 + [(22.5 − 10) / 16] × 10 = 70.5 + 7.81 = **78.31 marks**

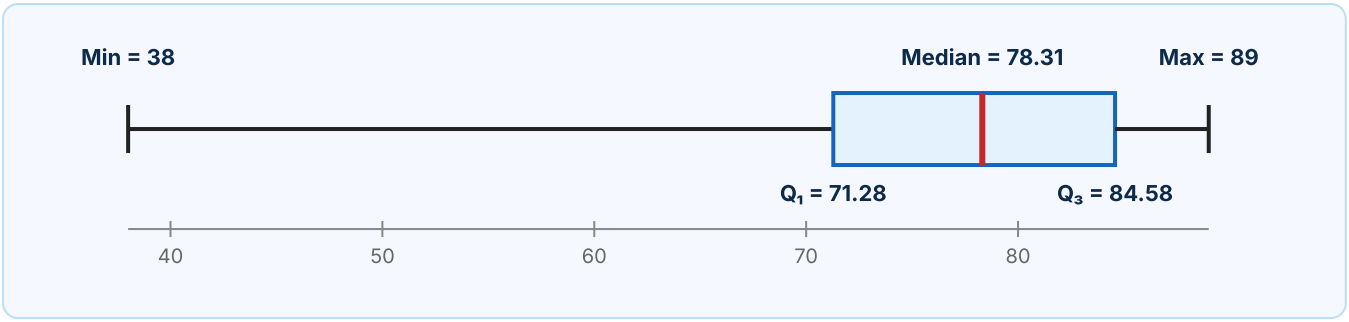
Mode: highest f = 19 → **modal class 81 – 90** (boundary 80.5 – 90.5).
L = 80.5, f_1 = 19, f_o = 16, f_2 = 0, c = 10
Mode = 80.5 + [(19 − 16) / (2×19 − 16 − 0)] × 10 = 80.5 + 1.36 = **81.86 marks**

(c)

Q_1: N/4 = 11.25 → class 71 – 80 (L = 70.5, cf = 10, f = 16, c = 10)
Q_1 = 70.5 + [(11.25 − 10) / 16] × 10 = 70.5 + 0.78 = **71.28 marks**

Q_3: 3N/4 = 33.75 → class 81 – 90 (L = 80.5, cf = 26, f = 19, c = 10)
Q_3 = 80.5 + [(33.75 − 26) / 19] × 10 = 80.5 + 4.08 = **84.58 marks**

(d) Five-number summary: Min = **38**, Q_1 = **71.28**, Median = **78.31**, Q_3 = **84.58**, Max = **89**. IQR = 13.3.



(e) Mean = 75.72, Median = 78.31, Mode = 81.86 → **Mean < Median < Mode**. In the box plot the left whisker is much longer than the right (left 33.28, right 4.42) and the median line lies closer to Q_3 (left half of the box is wider). Therefore the distribution is **negatively (left) skewed**.

Exam Practice 3 — Solution

(a) Range = 200 − 111 = 89. Classes of width 20 starting 101–120; last class 181–200 contains the maximum 200. Boundaries = lower − 0.5 and upper + 0.5; midpoint = (lower + upper)/2.

Class limits	Class boundaries	Tally	f	Midpoint x	c.f.
101 – 120	100.5 – 120.5	|	1	110.5	1
121 – 140	120.5 – 140.5	𝍸 |||	8	130.5	9
141 – 160	140.5 – 160.5	𝍸 𝍸 𝍸 |||	18	150.5	27
161 – 180	160.5 – 180.5	𝍸 𝍸 𝍸	15	170.5	42
181 – 200	180.5 – 200.5	𝍸 |	6	190.5	48
Total			**N = 48**		

Check: Σf = 48 = number of observations ✓

(b)

Class	f	x	fx
101 – 120	1	110.5	110.5
121 – 140	8	130.5	1044
141 – 160	18	150.5	2709
161 – 180	15	170.5	2557.5
181 – 200	6	190.5	1143
Total	**48**		**7564**

Mean = Σfx / N = 7564 / 48 = **157.58 bytes**

Median: N/2 = 24 → first c.f. ≥ 24 is 27 → **median class 141 – 160** (boundary 140.5 – 160.5).
L = 140.5, cf = 9, f = 18, c = 20
Median = 140.5 + [(24 − 9) / 18] × 20 = 140.5 + 16.67 = **157.17 bytes**

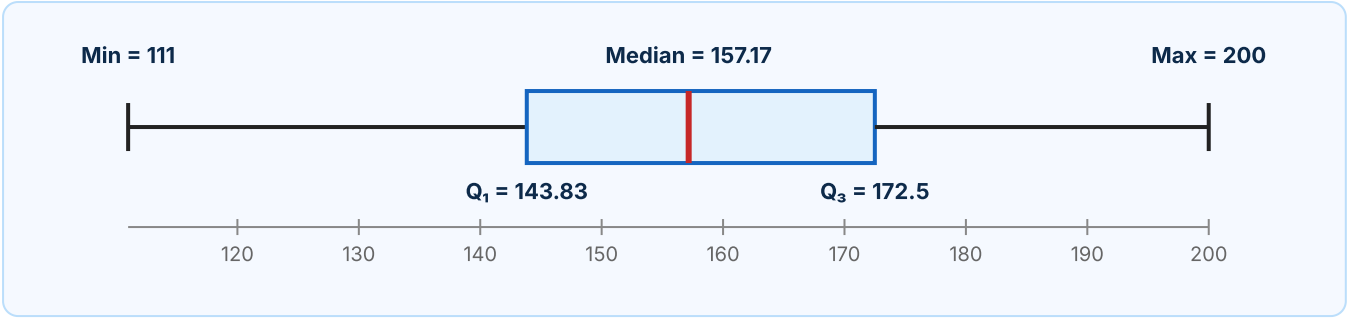
Mode: highest f = 18 → **modal class 141 – 160** (boundary 140.5 – 160.5).
L = 140.5, f_1 = 18, f_o = 8, f_2 = 15, c = 20
Mode = 140.5 + [(18 − 8) / (2×18 − 8 − 15)] × 20 = 140.5 + 15.38 = **155.88 bytes**

(c)

Q_1: N/4 = 12 → class 141 – 160 (L = 140.5, cf = 9, f = 18, c = 20)
Q_1 = 140.5 + [(12 − 9) / 18] × 20 = 140.5 + 3.33 = **143.83 bytes**

Q_3: 3N/4 = 36 → class 161 – 180 (L = 160.5, cf = 27, f = 15, c = 20)
Q_3 = 160.5 + [(36 − 27) / 15] × 20 = 160.5 + 12 = **172.5 bytes**

(d) Five-number summary: Min = **111**, Q_1 = **143.83**, Median = **157.17**, Q_3 = **172.5**, Max = **200**. IQR = 28.67.



(e) Mean = 157.58, Median = 157.17, Mode = 155.88 → **Mean ≈ Median ≈ Mode (they differ by less than 1.7)**. In the box plot the two whiskers are of similar length (left 32.83, right 27.5) and the median line lies closer to Q_1 (right half of the box is wider). Therefore the distribution is **approximately symmetric**.

Exam Practice 4 — Solution

(a) Range = 47 − 11 = 36. Classes of width 10 starting 11–20; last class 41–50 contains the maximum 47. Boundaries = lower − 0.5 and upper + 0.5; midpoint = (lower + upper)/2.

Class limits	Class boundaries	Tally	f	Midpoint x	c.f.
11 – 20	10.5 – 20.5	𝍸 𝍸 𝍸 𝍸	20	15.5	20
21 – 30	20.5 – 30.5	𝍸 𝍸	10	25.5	30
31 – 40	30.5 – 40.5	𝍸 ||	7	35.5	37
41 – 50	40.5 – 50.5	|||	3	45.5	40
Total			**N = 40**		

Check: Σf = 40 = number of observations ✓

(b)

Class	f	x	fx
11 – 20	20	15.5	310
21 – 30	10	25.5	255
31 – 40	7	35.5	248.5
41 – 50	3	45.5	136.5
Total	**40**		**950**

Mean = Σfx / N = 950 / 40 = **23.75 Mbps**

Median: N/2 = 20 → first c.f. ≥ 20 is 20 → **median class 11 – 20** (boundary 10.5 – 20.5).
L = 10.5, cf = 0, f = 20, c = 10
Median = 10.5 + [(20 − 0) / 20] × 10 = 10.5 + 10 = **20.5 Mbps**

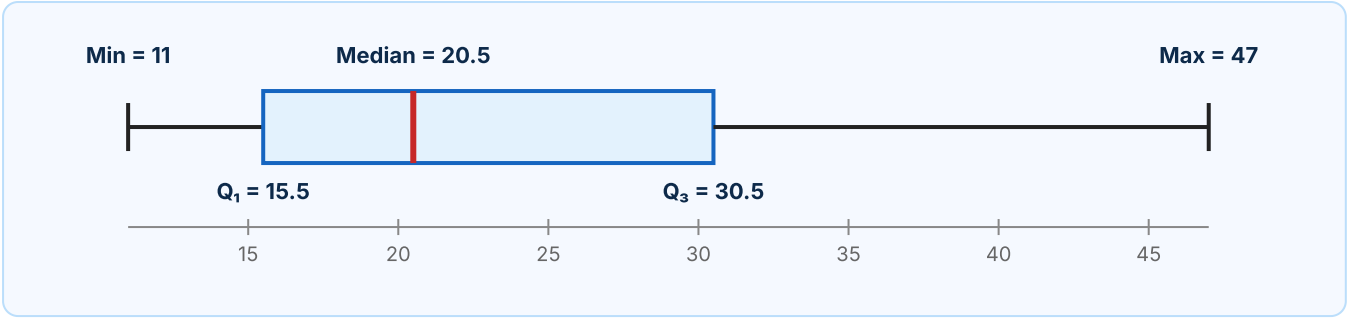
Mode: highest f = 20 → **modal class 11 – 20** (boundary 10.5 – 20.5).
L = 10.5, f_1 = 20, f_o = 0 (no class before), f_2 = 10, c = 10
Mode = 10.5 + [(20 − 0) / (2×20 − 0 − 10)] × 10 = 10.5 + 6.67 = **17.17 Mbps**

(c)

Q_1: N/4 = 10 → class 11 – 20 (L = 10.5, cf = 0, f = 20, c = 10)
Q_1 = 10.5 + [(10 − 0) / 20] × 10 = 10.5 + 5 = **15.5 Mbps**

Q_3: 3N/4 = 30 → class 21 – 30 (L = 20.5, cf = 20, f = 10, c = 10)
Q_3 = 20.5 + [(30 − 20) / 10] × 10 = 20.5 + 10 = **30.5 Mbps**

(d) Five-number summary: Min = **11**, Q_1 = **15.5**, Median = **20.5**, Q_3 = **30.5**, Max = **47**. IQR = 15.



(e) Mean = 23.75, Median = 20.5, Mode = 17.17 → **Mean > Median > Mode**. In the box plot the right whisker is much longer than the left (left 4.5, right 16.5) and the median line lies closer to Q_1 (right half of the box is wider). Therefore the distribution is **positively (right) skewed**.

Exam Practice 5 — Solution

(a) Range = 49 − 25 = 24. Classes of width 5 starting 21–25; last class 46–50 contains the maximum 49. Boundaries = lower − 0.5 and upper + 0.5; midpoint = (lower + upper)/2.

Class limits	Class boundaries	Tally	f	Midpoint x	c.f.
21 – 25	20.5 – 25.5	|	1	23	1
26 – 30	25.5 – 30.5	|	1	28	2
31 – 35	30.5 – 35.5	|	1	33	3
36 – 40	35.5 – 40.5	𝍸 𝍸	10	38	13
41 – 45	40.5 – 45.5	𝍸 𝍸 𝍸 |||	18	43	31
46 – 50	45.5 – 50.5	𝍸 𝍸 𝍸 ||||	19	48	50
Total			**N = 50**		

Check: Σf = 50 = number of observations ✓

(b)

Class	f	x	fx
21 – 25	1	23	23
26 – 30	1	28	28
31 – 35	1	33	33
36 – 40	10	38	380
41 – 45	18	43	774
46 – 50	19	48	912
Total	**50**		**2150**

Mean = Σfx / N = 2150 / 50 = **43 hours**

Median: N/2 = 25 → first c.f. ≥ 25 is 31 → **median class 41 – 45** (boundary 40.5 – 45.5).
L = 40.5, cf = 13, f = 18, c = 5
Median = 40.5 + [(25 − 13) / 18] × 5 = 40.5 + 3.33 = **43.83 hours**

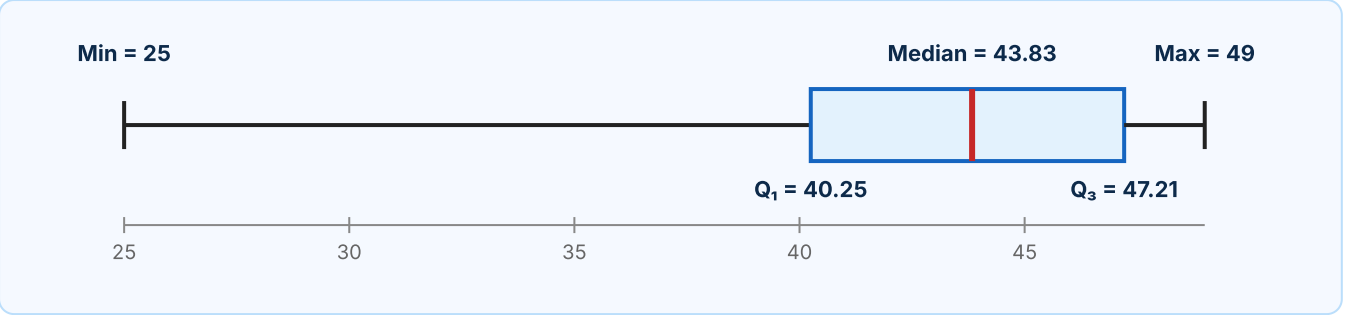
Mode: highest f = 19 → **modal class 46 – 50** (boundary 45.5 – 50.5).
L = 45.5, f_1 = 19, f_o = 18, f_2 = 0, c = 5
Mode = 45.5 + [(19 − 18) / (2×19 − 18 − 0)] × 5 = 45.5 + 0.25 = **45.75 hours**

(c)

Q_1: N/4 = 12.5 → class 36 – 40 (L = 35.5, cf = 3, f = 10, c = 5)
Q_1 = 35.5 + [(12.5 − 3) / 10] × 5 = 35.5 + 4.75 = **40.25 hours**

Q_3: 3N/4 = 37.5 → class 46 – 50 (L = 45.5, cf = 31, f = 19, c = 5)
Q_3 = 45.5 + [(37.5 − 31) / 19] × 5 = 45.5 + 1.71 = **47.21 hours**

(d) Five-number summary: Min = **25**, Q_1 = **40.25**, Median = **43.83**, Q_3 = **47.21**, Max = **49**. IQR = 6.96.



(e) Mean = 43, Median = 43.83, Mode = 45.75 → **Mean < Median < Mode**. In the box plot the left whisker is much longer than the right (left 15.25, right 1.79) and the median line lies closer to Q_3 (left half of the box is wider). Therefore the distribution is **negatively (left) skewed**.

Exam Practice 6 — Solution

(a) Range = 39 − 3 = 36. Classes of width 10 starting 1–10; last class 31–40 contains the maximum 39. Boundaries = lower − 0.5 and upper + 0.5; midpoint = (lower + upper)/2.

Class limits	Class boundaries	Tally	f	Midpoint x	c.f.
1 – 10	0.5 – 10.5	𝍸 𝍸 𝍸 ||||	19	5.5	19
11 – 20	10.5 – 20.5	𝍸 𝍸 𝍸 ||	17	15.5	36
21 – 30	20.5 – 30.5	𝍸 |||	8	25.5	44
31 – 40	30.5 – 40.5	|	1	35.5	45
Total			**N = 45**		

Check: Σf = 45 = number of observations ✓

(b)

Class	f	x	fx
1 – 10	19	5.5	104.5
11 – 20	17	15.5	263.5
21 – 30	8	25.5	204
31 – 40	1	35.5	35.5
Total	**45**		**607.5**

Mean = Σfx / N = 607.5 / 45 = **13.5 attempts**

Median: N/2 = 22.5 → first c.f. ≥ 22.5 is 36 → **median class 11 – 20** (boundary 10.5 – 20.5).
L = 10.5, cf = 19, f = 17, c = 10
Median = 10.5 + [(22.5 − 19) / 17] × 10 = 10.5 + 2.06 = **12.56 attempts**

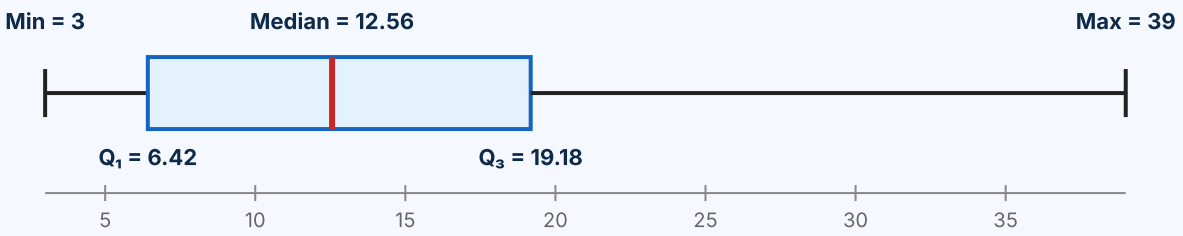
Mode: highest f = 19 → **modal class 1 – 10** (boundary 0.5 – 10.5).
L = 0.5, f_1 = 19, f_o = 0 (no class before), f_2 = 17, c = 10
Mode = 0.5 + [(19 − 0) / (2×19 − 0 − 17)] × 10 = 0.5 + 9.05 = **9.55 attempts**

(c)

Q_1: N/4 = 11.25 → class 1 – 10 (L = 0.5, cf = 0, f = 19, c = 10)
Q_1 = 0.5 + [(11.25 − 0) / 19] × 10 = 0.5 + 5.92 = **6.42 attempts**

Q_3: 3N/4 = 33.75 → class 11 – 20 (L = 10.5, cf = 19, f = 17, c = 10)
Q_3 = 10.5 + [(33.75 − 19) / 17] × 10 = 10.5 + 8.68 = **19.18 attempts**

(d) Five-number summary: Min = **3**, Q_1 = **6.42**, Median = **12.56**, Q_3 = **19.18**, Max = **39**. IQR = 12.76.



(e) Mean = 13.5, Median = 12.56, Mode = 9.55 → **Mean > Median > Mode**. In the box plot the right whisker is much longer than the left (left 3.42, right 19.82) and the median line lies closer to Q_1 (right half of the box is wider). Therefore the distribution is **positively (right) skewed**.

Exam Practice 7 — Solution

(a) Range = 86 − 48 = 38. Classes of width 10 starting 41–50; last class 81–90 contains the maximum 86. Boundaries = lower − 0.5 and upper + 0.5; midpoint = (lower + upper)/2.

Class limits	Class boundaries	Tally	f	Midpoint x	c.f.
41 – 50	40.5 – 50.5	||	2	45.5	2
51 – 60	50.5 – 60.5	𝍸 ||	7	55.5	9
61 – 70	60.5 – 70.5	𝍸 𝍸 𝍸 |	16	65.5	25
71 – 80	70.5 – 80.5	𝍸 𝍸 |	11	75.5	36
81 – 90	80.5 – 90.5	||||	4	85.5	40
Total			**N = 40**		

Check: Σf = 40 = number of observations ✓

(b)

Class	f	x	fx
41 – 50	2	45.5	91
51 – 60	7	55.5	388.5
61 – 70	16	65.5	1048
71 – 80	11	75.5	830.5
81 – 90	4	85.5	342
Total	**40**		**2700**

Mean = Σfx / N = 2700 / 40 = **67.5 min**

Median: N/2 = 20 → first c.f. ≥ 20 is 25 → **median class 61 – 70** (boundary 60.5 – 70.5).
L = 60.5, cf = 9, f = 16, c = 10
Median = 60.5 + [(20 − 9) / 16] × 10 = 60.5 + 6.88 = **67.38 min**

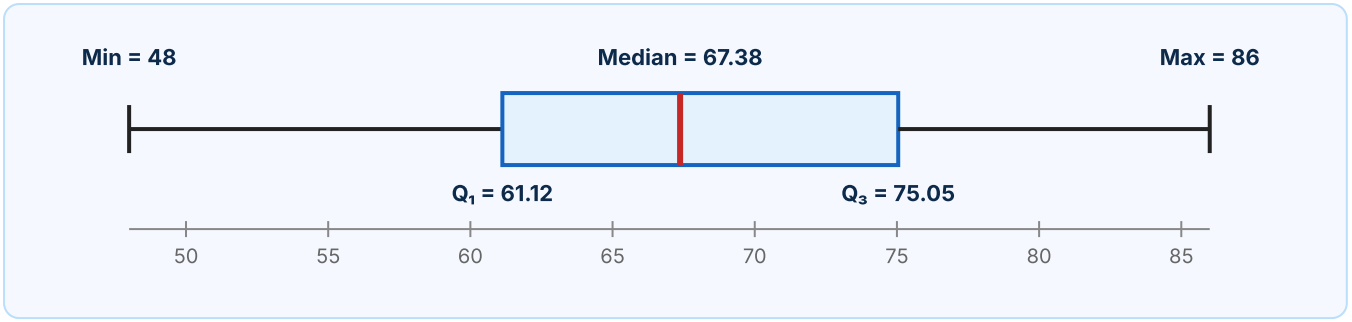
Mode: highest f = 16 → **modal class 61 – 70** (boundary 60.5 – 70.5).
L = 60.5, f_1 = 16, f_o = 7, f_2 = 11, c = 10
Mode = 60.5 + [(16 − 7) / (2×16 − 7 − 11)] × 10 = 60.5 + 6.43 = **66.93 min**

(c)

Q_1: N/4 = 10 → class 61 – 70 (L = 60.5, cf = 9, f = 16, c = 10)
Q_1 = 60.5 + [(10 − 9) / 16] × 10 = 60.5 + 0.62 = **61.12 min**

Q_3: 3N/4 = 30 → class 71 – 80 (L = 70.5, cf = 25, f = 11, c = 10)
Q_3 = 70.5 + [(30 − 25) / 11] × 10 = 70.5 + 4.55 = **75.05 min**

(d) Five-number summary: Min = **48**, Q_1 = **61.12**, Median = **67.38**, Q_3 = **75.05**, Max = **86**. IQR = 13.92.



(e) Mean = 67.5, Median = 67.38, Mode = 66.93 → **Mean ≈ Median ≈ Mode (they differ by less than 0.58)**. In the box plot the two whiskers are of similar length (left 13.12, right 10.95) and the median line lies closer to Q_1 (right half of the box is wider). Therefore the distribution is **approximately symmetric**.

Exam Practice 8 — Solution

(a) Range = 206 − 183 = 23. Classes of width 5 starting 181–185; last class 206–210 contains the maximum 206. Boundaries = lower − 0.5 and upper + 0.5; midpoint = (lower + upper)/2.

Class limits	Class boundaries	Tally	f	Midpoint x	c.f.
181 – 185	180.5 – 185.5	|	1	183	1
186 – 190	185.5 – 190.5	𝍸 |||	8	188	9
191 – 195	190.5 – 195.5	𝍸 𝍸 𝍸 ||||	19	193	28
196 – 200	195.5 – 200.5	𝍸 𝍸 𝍸 ||	17	198	45
201 – 205	200.5 – 205.5	|||	3	203	48
206 – 210	205.5 – 210.5	||	2	208	50
Total			**N = 50**		

Check: Σf = 50 = number of observations ✓

(b)

Class	f	x	fx
181 – 185	1	183	183
186 – 190	8	188	1504
191 – 195	19	193	3667
196 – 200	17	198	3366
201 – 205	3	203	609
206 – 210	2	208	416
Total	**50**		**9745**

Mean = Σfx / N = 9745 / 50 = **194.9 g**

Median: N/2 = 25 → first c.f. ≥ 25 is 28 → **median class 191 – 195** (boundary 190.5 – 195.5).
L = 190.5, cf = 9, f = 19, c = 5

Median = 190.5 + [(25 − 9) / 19] × 5 = 190.5 + 4.21 = **194.71 g**

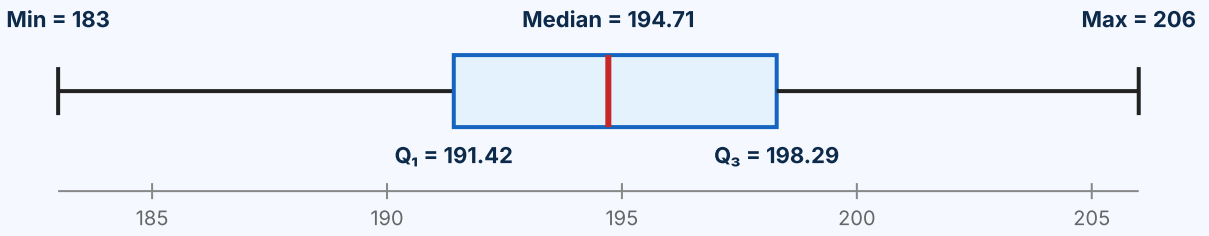
Mode: highest f = 19 → **modal class 191 – 195** (boundary 190.5 – 195.5).
L = 190.5, f_1 = 19, f_o = 8, f_2 = 17, c = 5
Mode = 190.5 + [(19 − 8) / (2×19 − 8 − 17)] × 5 = 190.5 + 4.23 = **194.73 g**

(c)

Q_1: N/4 = 12.5 → class 191 – 195 (L = 190.5, cf = 9, f = 19, c = 5)
Q_1 = 190.5 + [(12.5 − 9) / 19] × 5 = 190.5 + 0.92 = **191.42 g**

Q_3: 3N/4 = 37.5 → class 196 – 200 (L = 195.5, cf = 28, f = 17, c = 5)
Q_3 = 195.5 + [(37.5 − 28) / 17] × 5 = 195.5 + 2.79 = **198.29 g**

(d) Five-number summary: Min = **183**, Q_1 = **191.42**, Median = **194.71**, Q_3 = **198.29**, Max = **206**. IQR = 6.87.



(e) Mean = 194.9, Median = 194.71, Mode = 194.73 → **Mean ≈ Median ≈ Mode (they differ by less than 0.17)**. In the box plot the two whiskers are of similar length (left 8.42, right 7.71) and the median line lies closer to Q_1 (right half of the box is wider). Therefore the distribution is **approximately symmetric**.

Exam Practice 9 — Solution

(a) Range = 99 − 47 = 52. Classes of width 10 starting 41–50; last class 91–100 contains the maximum 99. Boundaries = lower − 0.5 and upper + 0.5; midpoint = (lower + upper)/2.

Class limits	Class boundaries	Tally	f	Midpoint x	c.f.
41 – 50	40.5 – 50.5	|	1	45.5	1
51 – 60	50.5 – 60.5	|	1	55.5	2
61 – 70	60.5 – 70.5	||||	4	65.5	6
71 – 80	70.5 – 80.5	𝍸 ||||	9	75.5	15
81 – 90	80.5 – 90.5	𝍸 𝍸 𝍸 |||	18	85.5	33
91 – 100	90.5 – 100.5	𝍸 𝍸 𝍸	15	95.5	48
Total			**N = 48**		

Check: Σf = 48 = number of observations ✓

(b)

Class	f	x	fx
41 – 50	1	45.5	45.5
51 – 60	1	55.5	55.5
61 – 70	4	65.5	262
71 – 80	9	75.5	679.5
81 – 90	18	85.5	1539
91 – 100	15	95.5	1432.5

Total	48		4014

Mean = Σfx / N = 4014 / 48 = **83.62 °C**

Median: N/2 = 24 → first c.f. ≥ 24 is 33 → **median class 81 – 90** (boundary 80.5 – 90.5).
L = 80.5, cf = 15, f = 18, c = 10
Median = 80.5 + [(24 − 15) / 18] × 10 = 80.5 + 5 = **85.5 °C**

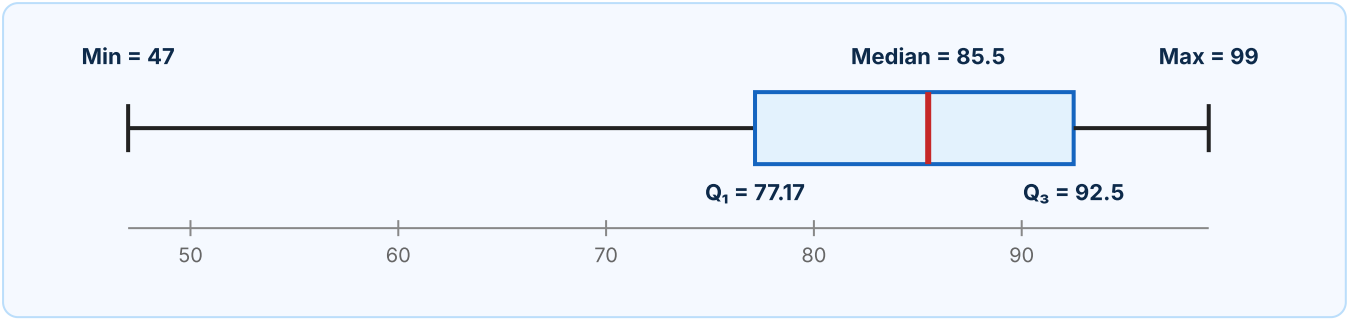
Mode: highest f = 18 → **modal class 81 – 90** (boundary 80.5 – 90.5).
L = 80.5, f_1 = 18, f_o = 9, f_2 = 15, c = 10
Mode = 80.5 + [(18 − 9) / (2×18 − 9 − 15)] × 10 = 80.5 + 7.5 = **88 °C**

(c)

Q_1: N/4 = 12 → class 71 – 80 (L = 70.5, cf = 6, f = 9, c = 10)
Q_1 = 70.5 + [(12 − 6) / 9] × 10 = 70.5 + 6.67 = **77.17 °C**

Q_3: 3N/4 = 36 → class 91 – 100 (L = 90.5, cf = 33, f = 15, c = 10)
Q_3 = 90.5 + [(36 − 33) / 15] × 10 = 90.5 + 2 = **92.5 °C**

(d) Five-number summary: Min = **47**, Q_1 = **77.17**, Median = **85.5**, Q_3 = **92.5**, Max = **99**. IQR = 15.33.



(e) Mean = 83.62, Median = 85.5, Mode = 88 → **Mean < Median < Mode**. In the box plot the left whisker is much longer than the right (left 30.17, right 6.5) and the median line lies closer to Q_3 (left half of the box is wider). Therefore the distribution is **negatively (left) skewed**.

Exam Practice 10 — Solution

(a) Range = 99 − 31 = 68. Classes of width 20 starting 31–50; last class 91–110 contains the maximum 99. Boundaries = lower − 0.5 and upper + 0.5; midpoint = (lower + upper)/2.

Class limits	Class boundaries	Tally	f	Midpoint x	c.f.
31 – 50	30.5 – 50.5	𝍸 𝍸 𝍸 𝍸 ||	22	40.5	22
51 – 70	50.5 – 70.5	𝍸 𝍸 𝍸 |||	18	60.5	40
71 – 90	70.5 – 90.5	𝍸 ||	7	80.5	47
91 – 110	90.5 – 110.5	|||	3	100.5	50
Total			**N = 50**		

Check: Σf = 50 = number of observations ✓

(b)

Class	f	x	fx
31 – 50	22	40.5	891
51 – 70	18	60.5	1089
71 – 90	7	80.5	563.5
91 – 110	3	100.5	301.5

Total	**50**		**2845**

Mean = Σfx / N = 2845 / 50 = **56.9 ms**

Median: N/2 = 25 → first c.f. ≥ 25 is 40 → **median class 51 – 70** (boundary 50.5 – 70.5).
L = 50.5, cf = 22, f = 18, c = 20
Median = 50.5 + [(25 − 22) / 18] × 20 = 50.5 + 3.33 = **53.83 ms**

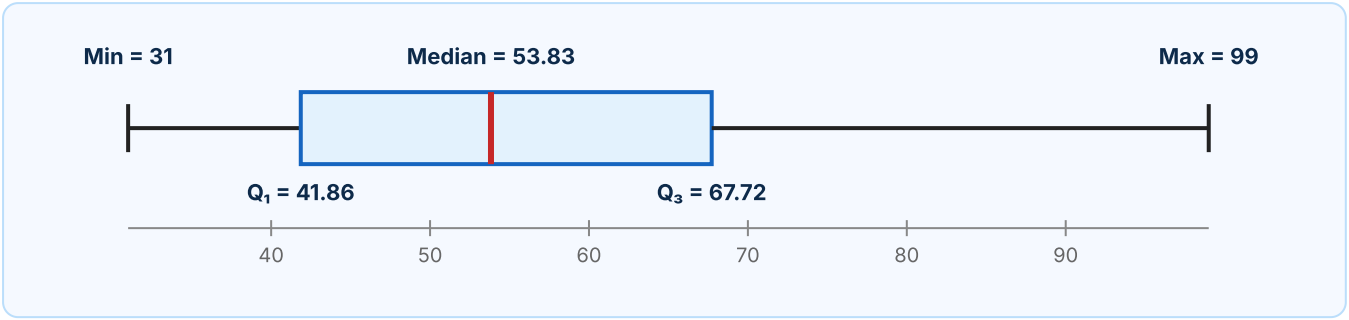
Mode: highest f = 22 → **modal class 31 – 50** (boundary 30.5 – 50.5).
L = 30.5, f_1 = 22, f_o = 0 (no class before), f_2 = 18, c = 20
Mode = 30.5 + [(22 − 0) / (2×22 − 0 − 18)] × 20 = 30.5 + 16.92 = **47.42 ms**

(c)

Q_1: N/4 = 12.5 → class 31 – 50 (L = 30.5, cf = 0, f = 22, c = 20)
Q_1 = 30.5 + [(12.5 − 0) / 22] × 20 = 30.5 + 11.36 = **41.86 ms**

Q_3: 3N/4 = 37.5 → class 51 – 70 (L = 50.5, cf = 22, f = 18, c = 20)
Q_3 = 50.5 + [(37.5 − 22) / 18] × 20 = 50.5 + 17.22 = **67.72 ms**

(d) Five-number summary: Min = **31**, Q_1 = **41.86**, Median = **53.83**, Q_3 = **67.72**, Max = **99**. IQR = 25.86.



(e) Mean = 56.9, Median = 53.83, Mode = 47.42 → **Mean > Median > Mode**. In the box plot the right whisker is much longer than the left (left 10.86, right 31.28) and the median line lies closer to Q_1 (right half of the box is wider). Therefore the distribution is **positively (right) skewed**.